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RESEARCH ARTICLE

A Minimalistic Distance-Vector Routing Protocol for LoRa Mesh Networks

ROGER PUEYO CENTELLES, ROC MESEGUER, FELIX FREITAG, (Member, IEEE), ROGER BAIG VIÑAS, AND LEANDRO NAVARRO

Computer Architecture Department, Universitat Politècnica de Catalunya (BarcelonaTech), 08034 Barcelona, Spain

Corresponding author: Felix Freitag (felix.freitag@upc.edu)

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ABSTRACT LoRa is popular in the Internet of Things (IoT) domain as a Low Power, Wide Area Network (LPWAN) radio technology, providing low-power and long-range communication in the sub-GHz band. Most often, LoRa is used as part of the LoRaWAN architecture with a star-of-stars topology, but it can also be operated standalone. This paper presents and evaluates a minimalistic Routing Protocol (RP) for building LoRa networks with a more flexible mesh network topology. We propose a Time on Air (ToA) metric that, when used in heterogeneous network topologies, can take better advantage of LoRa's multiple Spreading Factors (SFs), and their trade-off between transmission distance and bit rate, and their quasi-orthogonality property. We evaluate the routing protocol with the FLoRa framework and OMNeT++ and compare it with other common routing strategies. Our experiments provide a comprehensive understanding of the routing protocol performance concerning the scalability, throughput, and latency in several topology and network traffic scenarios. When using the ToA metric, we observed in LoRa mesh networks formed by random topologies and heterogeneous links the positive impact of a more balanced Packet Delivery Ratio (PDR) among the nodes with different network centrality, suggesting hence the multi-SF ToA metric to be used for improved PDR in conditions that are expected to occur in real-world systems. Regarding goodput, we observed that using multiple SF simultaneously had a limited impact on low to high traffic loads but dramatically improved throughput and goodput in traffic saturation scenarios. This suggests that multi-SF operation would be preferred to maximize network performance. We observed the best latency performance with multi-SF ToA for low and medium traffic loads, especially for the random topology, where the metric can again take advantage of the node and link heterogeneity.

INDEX TERMS LoRa, mesh network, routing.

I. INTRODUCTION

Several technologies have emerged during the last decade providing wireless communication for IoT devices in various scenarios [1]. Among them, LoRa has proven successful in the LPWAN domain for a vast number of deployments in diverse environments that require the transmission of small amounts of data over long distances (Figure 1).

LoRaWAN is an open specification by the LoRa Alliance that provides the Medium Access Control (MAC) and application layers on top of the LoRa radio technology, defining a star of stars topology with gateways at the center and end nodes around them [2]. This network architecture has proven suitable for many diverse IoT applications [3]. However, there are reasons for using a more flexible topology than LoRaWAN's single-hop, gateway-client scheme. For instance, to increase network coverage, to deal with challenging geography, to overcome the need for additional infrastructure besides the end devices themselves in Peer-To-Peer (P2P) communications, or even to avoid data going to the cloud through an Internet connection in underserved locations [4].

Figure 2 depicts the LoRaWAN architecture, signaling the issues mentioned above. Our proposal, rather than extending LoRaWAN's set of features, aims at breaking the hierarchical star of stars topology and contributing to

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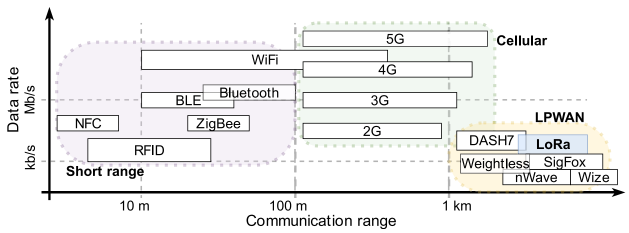


FIGURE 1. A non-exhaustive classification of radio technologies and protocols for IoT according to their approximate communication range and data rate. LoRa provides long-range links, of up to a few km, at tens of kbps.

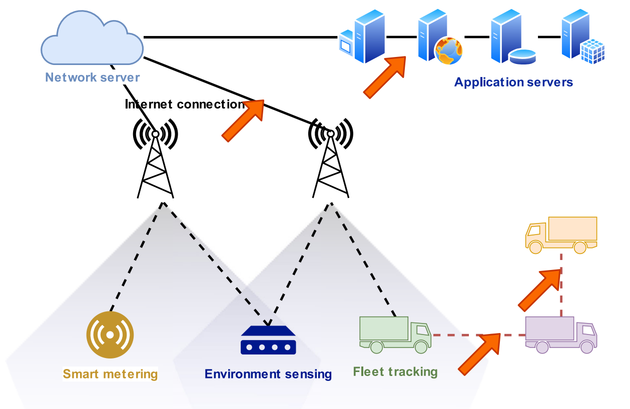


FIGURE 2. A schematic representation of the LoRaWAN architecture. The gateways in the center define the star of stars topology and the coverage area for the end nodes at the bottom. At the top, the network and application servers usually reside in the cloud. The orange arrows indicate the main issues LoRa mesh networking can address: dependence on the cloud, Internet connectivity, gateway infrastructure, and P2P communication between end nodes.

building LoRa mesh networks that allow communication between any pair of nodes without the need for the gateway infrastructure [5] increasing the service of a LoRa mesh network. However, mesh topologies come at the price of requiring the nodes to spend resources (CPU, memory, power, airtime) to maintain the network operative and forward data from other participants in the network that could not otherwise communicate.

This paper presents a minimalistic Distance-vector (DV) Routing Protocol (RP) built on the multi-hop LoRa mesh networks we developed in [6]. We focus on the design and performance comparison of the novel Time on Air (ToA) routing metric. The RP leverages the Spreading Factor (SF). This modulation parameter balances communication range and bit rate, providing multiple channels of quasi-orthogonal signals when different SF values are used with single radio hardware. This setup allows simultaneous transmissions on the same frequency under certain conditions. The ToA metric utilizes the multi-SF characteristics of LoRa to determine the best route between nodes based on the total transmission time required along the entire path, making it sensitive to congestion.

To evaluate the RP and the different routing metrics, including ToA, we use the FLoRa framework [7] and the OMNeT++ simulator [8], tested in different mesh network deployments with up to 64 nodes. We analyze key performance aspects such as PDR, throughput, and latency under different setups and compare our proposed metric with other well-known routing strategies.

The following Section II provides a brief introduction to the LoRa radio technology. Routing for LoRa –and most often, for LoRaWAN– is an active research topic; we selected the most relevant state-of-the-art in Section III. In Section IV, we introduce our minimalist DV RP for building LoRa mesh networks and describe its fundamentals, including the ToA metric used to evaluate the cost of paths to the nodes. Section V discusses the methodology followed in the experiments, explaining the simulation framework, the network topologies, the nodes' characteristics, etc., and includes a benchmark comparing the simulator performance with real hardware. Section VI evaluates the simulation experiments performed, analyzing the capabilities that multi-SF-aware routing for LoRa adds and comparing the ToA metric with other well-known routing strategies. We summarize the most remarkable findings in Section VII, including a look at the open issues and future work.

II. THE LoRa RADIO TECHNOLOGY

LoRa, which stands for long range, is a wireless communication technology owned by Semtech[1] that operates in the sub-gigahertz range of the radio spectrum. It employs Chirp Spread Spectrum (CSS), a proprietary modulation technique resistant to multi-path fading and suitable for noisy environments, aiming to provide low throughput communication with links of more than 10 km –outdoors, in rural areas– while maintaining low power consumption.

Several parameters of the LoRa physical layer can be configured to optimize communications for a given scenario or application: radio band and frequency, channel bandwidth, transmission power, Forward Error Correction (FEC) rate, and SF. These settings can be applied network-wide or on a per-device basis. IoT deployments commonly operate on license-exempt Industrial, Scientific, Medical (ISM) bands, which change from one geographic area to another; diverse LoRa transceivers can operate in any of them. Many channels are available inside these bands for up-link and down-link transmissions, with thinner or narrower channel bandwidths and different maximum transmission powers allowed. In addition to the robust CSS modulation, LoRa's FEC protects against interference on noisy links. Table 1 summarizes the configurable parameters and lists their possible values.

The SF is, perhaps, the most distinctive configuration parameter in LoRa, as it determines a direct trade-off between communication range and *data rate*. Furthermore, two LoRa transmissions on the same frequency using different SFs are quasi-orthogonal [9] meaning that, generally, both can be successfully demodulated simultaneously, each by a different receiver. This feature is leveraged by LoRa gateway

[1] Semtech LoRa Technology Overview - https://www.semtech.com/lora

TABLE 1. Configurable parameters in LoRa transmissions.

Configurable parameter	Values
Radio band	169, 433, 868, 915 MHz [a]
Bandwidth	62.5, 125, 250, 500 kHz [b]
Transmission power	14 dBm (EU), 27 dBm (USA)
Spreading Factor	6 to 12 [c]
FEC rate	4/5, 4/6, 4/7, 4/8

[a] Common frequency allocations for ISM in different regions worldwide; LoRa may also be used in licensed bands.
[b] Smaller bandwidths (7.8 to 41.7 kHz) are also supported, although rarely used.
[c] Certain bandwidth and SF combinations may result in too long transmissions for specific radio bands in which duty cycle or time-on-air limitations often apply.

transceiver chips, extensively used in the LoRaWAN architecture. Gateway chips include different signal sampling, decimation, switching, and processing steps in their die. They are capable of receiving up to nine LoRa transmissions concurrently[2] (plus a tenth one using other modulations). End nodes typically feature a single-channel transceiver with a more straightforward design and a much-reduced cost. These chips allow only half-duplex transmission/reception on a single channel and with a single SF at a time.

A drawback for LoRa (and, in fact, for any radio technology operating in the sub- GHz part of the spectrum) is the legal duty cycle limitation imposed in ISM bands, which only allows a device to transmit on a given channel for a maximum of percentage of time (e.g., in Europe, duty cycle in the 868 MHz ISM band is 1 %). Sometimes, depending on the local regulation, this limitation can be relaxed if Channel Activity Detection (CAD) or Listen-before-talk (LBT) mechanisms are implemented.

III. STATE-OF-THE-ART

Several proposals regarding multi-hop, mesh, and routing for LoRa and LoRaWAN have emerged recently. They have been thoroughly classified and analyzed by different researchers from different points of view: taking the application scenarios into account [4], focusing on the LoRaWAN architecture [10], or on specific implementation aspects like topology and routing [11]. Their maturity and Technology Readiness Levels (TRLs) are heterogeneous and range from theoretical contributions to experimentally validated proposals in testbed or real-world deployments.

The communication range and transmission rate become significant factors influencing deployment costs as the system scales up. The scalability of LoRa nodes at a low cost depends on the ability of transceivers to cover a large area [12] reliably. This fact presents a challenge as it raises the overall costs and efforts required for expanding the coverage range of LoRa networks.

[2] Semtech SX1301 Digital Baseband Chip - https://www.semtech.com/products/wireless-rf/lora-core/sx1301

In this section, we analyze the most relevant proposals, classifying them depending on their application domain, the scenario they tackle, or the specific features they provide.

A. MULTI-HOP AND MESH ON TOP OF THE LoRaWAN ARCHITECTURE

The extension of a LoRaWAN network is determined by the coverage of the gateway –or gateways– that form it. Even if LoRa allows transmitting data over several km, sometimes this is not enough to cover vast areas, especially when deploying more gateways is technically or economically not feasible. Instead, adding multi-hop capability on top of the LoRaWAN architecture has been proposed, allowing packets to travel through different devices until they reach their destination. Three main strategies are found: adding multi-hop to the gateways, adding it to the end nodes, or introducing intermediate relaying devices [13].

Tian et al. [14] leveraged Concurrent Transmissions (CTs) to build LoRaHop, a protocol with multi-hop support for LoRaWAN networks in uplink and downlink directions. Built upon LoRaDisC (the authors' protocol that provides regular end nodes with the capacity to form a mesh network), it allows for relaying messages between a gateway and the other nodes. To this end, it performs different flooding rounds to disseminate data using CT and network coding, hence not requiring building and updating routing tables. Their protocol learns the predictable data transmission patterns of LoRaWAN end nodes to use the nodes' idle slots and reduce interference. Utilizing multi-hop, LoRaHop extends LoRaWAN coverage and allows using faster SFs with less power consumption.

Islam et al. [15] modified the Distance Ring Exponential Stations Generator (DRESG) framework. The relay operation uses intermediate gateways and employs a distance-based adaptive transmission configuration. The network topology is structured as a tree, allowing for flexible node positioning. Nodes are distributed into distance-rings within the system. End nodes are randomly deployed, with the primary gateway at the network center. A non-linear distance-spreading model establishes virtual rings around the main gateway. The end nodes are between successive rings, and each network node is associated with its nearest virtual ring. Subsequently, the system organizes multiple clusters, each comprising nodes with the same hop count required to reach the primary gateway. The formation of these clusters is determined by the distance of nodes from the main gateway. Intermediate gateways are strategically positioned within each ring, a placement contingent on the specific application scenario.

Lundell et al. [16] designed a routing protocol to provide mesh networking between gateways to extend coverage in urban and rural scenarios. This way, gateways without Internet access can forward packets to those with a backhaul connection. They adapted Hybrid Wireless Mesh Protocol (HWMP) and Ad-Hoc On-Demand Distance Vector (AODV) to the characteristics of LoRa and built a tunneling mechanism that operates transparently to both end nodes and the LoRaWAN server. The protocol was validated only with up-link messages in a 4-hops network with in-lab experiments.

Ebi et al. [17] implemented a synchronous LoRa mesh protocol to extend LoRaWAN networks for end nodes monitoring underground infrastructures. Their approach adds repeater nodes that bridge the synchronous LoRa mesh network segment with the regular LoRaWAN gateway. The results outperform a standard LoRaWAN network concerning the reliability of packet delivery when transmitting from range-critical locations, like underground areas. The solution enhances transmission reliability, efficiency, and flexibility but requires a precise time reference (e.g., using GPS or DCF77 time signaling) for synchronization.

B. MULTI-HOP, MESH, AND ROUTING FOR LoRa

In this section, our focus shifts toward complex multi-hop or mesh topologies. While several works propose linear topologies, as exemplified by works [18], [19], these do not necessitate routing as there is only one possible path between source and destination nodes.

Several proposals for multi-hop networks using LoRa do not belong to, or extend, the LoRaWAN architecture. Using alternative strategies like routing, Time-Division Multiple Access (TDMA), clustering techniques, etc., they create tree and mesh topologies to build more decentralized and flexible networks. Systems are often built with only single-channel radio nodes, but some combine them with multi-channel gateway hardware.

Sartori et al. [20] addressed the LoRaWAN coverage extension topic with RLMAC, a MAC layer protocol that enables Routing over Low Power and Lossy Networks (RPL) multi-hop communications based on LoRa. They argue that the star topology is convenient for ease of deployment and, from a business perspective. However, multi-hop could be the only option for covering vast areas with few base stations. Furthermore, it could mitigate congestion issues and increase throughput or reduce ToA by using faster SFs. The authors designed a multi-hop solution for single-channel LoRa nodes. They implemented the algorithms to bootstrap and operate a network using RPL by combining a slow reception loop with fast transmission loops. This approach ensures that nodes can receive messages using any SFs, albeit at a high synchronization cost.

Lee and Ke [21] designed and implemented a LoRa mesh networking system to ensure indoor nodes can communicate with network servers without deploying more gateways. Their design consists of a *data sink* (somewhat misleadingly, the authors call it a *gateway*) broadcasting beacons to invite nodes to join the network. Those, in turn, set the gateway (i.e., the data sink) as their parent. New nodes that hear packets from the gateway, or other nodes, can also join the network, choosing a suitable parent based on multiple factors (namely, Received Signal Strength Indicator (RSSI), hop count). The gateway polls children nodes to request their data and holds a complete view of the network topology, which it can modify based on its comprehensive information. The authors state that while their solution extends a network's coverage without installing more gateways, the number of serviced nodes would be smaller than with a conventional star topology because of the latency introduced by successive packet forwarding.

Zhu et al. [22] improved the capacity of a multi-hop LoRa network by off-loading traffic into several subnetworks with different SFs. This clustering technique results in a multiple-access dimension network where each subnetwork is rooted at a sink node with a specific SF. This technique makes packet transmission in parallel with multiple SFs feasible. Their solution ensures the connectivity of all subnetworks, off-loads traffic according to the number of nodes, data rates, and topologies, and shortens the ToA by reducing the hop count. The authors present a Tree-based SF Clustering Algorithm (TSCA) that conducts node allocation. Their solution requires a coordinated effort for the clustering decision-making tasks.

Mai and Kim [23] proposed a collision-free multi-hop LoRa network protocol with low latency. In their network, the sink node exchanges packets with the other nodes to construct a tree topology, assigning a timeslot and a channel to each link. This way, communication between the leaf and parent nodes is collision-free with the neighbors, as nodes transmit on their frequency during their assigned timeslot. The authors state that their protocol provides high reliability, parallel transmissions, low latency, and minimized timeslots and packet size. However, it is only suitable for networks with static topology where all the collected data are targeted towards a single sink node.

Prade et al. [24] introduced a multi-radio and multi-hop LoRa communication architecture to enhance the coverage and service for large-scale IoT deployment in rural areas, called Multi-LoRa. Their multi-hop architecture considered the limitations of LoRa deployment in scenarios like farms with hundreds of kilometers. They also presented a hardware prototype implementing their design, which improved the delay and packet loss figures compared to other setups in a physical testbed and a simulation environment.

Berto et al. [25] introduced a preliminary study to establish a LoRa-based mesh network. The prototype of this network is based on RadioHead. The message's header includes information necessary for routing and forwarding, whether the data is unreliable or reliable, through retransmission upon request. The network ensures multi-hop delivery of datagrams from a source node to a destination node, potentially involving zero or more intermediate nodes. The delivery is achieved through automatic route discovery and re-discovery, facilitated by a particular route discovery request broadcast packets. These packets are generated by the source node and conveyed to the destination node, utilizing a reactive routing approach. To validate their research, the authors presented a hardware/software prototype that employed low-power-consumption devices and provided an initial assessment of the proposed solution.

As detailed in their publication, Leonardi et al. [26] introduced MRT-LoRa, a multi-hop real-time communication protocol designed specifically for LoRa networks. This protocol adopts a tree topology organized into levels and employs Time-Division Multiple Access (TDMA) to fulfill its multi-hop and real-time communication objectives. It structures the network time into repeating superframes. Each superframe comprises a set of timeslots designated for various purposes, such as transmitting/receiving beacons, global acknowledgments, data transmission, or data forwarding. These timeslots are scheduled offline across channels and Spreading Factors (SFs). The critical distinction between this approach and our proposal lies in the offline scheduling mechanism.

Ghosh et al. [27] proposed LoRaute, where a LoRa mesh network is conceived as a backhaul network with hosts and mobile nodes. Routing nodes request their neighbor nodes for a list of their neighboring nodes. The obtained list forms the routing table of that node. While the experiments were done with actual nodes, the scale for evaluating the routing capacity was limited to at most seven nodes, and the results were tested with only a few scenarios, such that the scope of the results cannot easily be generalized.

Arratia et al. [28] presented a LoRa network-based solution to solve the data transmission of buoy sensor nodes in a lagoon. A multi-hop network forwarded messages without the need for a routing protocol. The evaluation was centered on practical aspects of the IoT application case, such as the throughput with different SFs and energy consumption. The evaluation scale was limited to a few nodes corresponding to the actual deployment of the network.

In [29], we introduced the LoRaMesher library. The LoRaMesher library offers multi-hop mesh delivery capabilities in the form of a real implementation that enables the operation of a LoRa mesh network on physical nodes. However, LoRaMesher's routing protocol is single-SF and uses a simple hop count metric, limiting certain scenarios' operation. Because LoRaMesher is not available as a simulator, it is not easy to employ a real deployment to conduct the performance evaluation of a large-scale design space. On the contrary, the simulation results of the multi-SF ToA of this paper can feed back into the implementation improvements of LoRaMesher.

Summarizing the above works, the most used routing metric in today's practical state-of-the-art LoRa mesh networks is the hop count in single SF scenarios. While the results demonstrate the feasibility of LoRa mesh networks, there is still a lack of understanding of more complex routing metrics for these networks and how the multi-SF potential that LoRa offers can be integrated into the network operation.

C. MULTI-SF RECEPTION WITH SINGLE-CHANNEL LoRa RADIO IN CLIENT NODES

As discussed in their work, Croce et al. [30] demonstrated that it is not always possible to achieve independent LoRa transmissions using different spreading factors simultaneously due to imperfect orthogonality. However, this limitation does not prevent the creation of links between different nodes within a mesh network using different Spreading Factors (SFs). This approach effectively expands the network's bandwidth by utilizing multiple SFs simultaneously and mitigates interference issues that may arise when the same SF is used nearby.

Westenberg [31] first exploited the single-node multi-SF reception strategy for building single-channel LoRaWAN gateways out of inexpensive ESP8266 and ESP32-based end node devices. This solution is not related to multi-hop or mesh but demonstrates that a single-channel radio can be used more flexibly. The LoRa chip is configured in a way that it detects the SF of an incoming transmission automatically, so the receiver is dynamically configured every time to decode the signal regardless of the SF used by the sender. Given the limited capabilities of the transceiver, the reception is still restricted to one LoRa packet at a time on a single channel. However, this strategy offers the flexibility of using any available SFs instead of a fixed predefined one.

Kim et al. [32] proposed an Adaptive Spreading Factor Selection (ASFS) scheme to build LoRa mesh networks using single-channel transceivers, increasing throughput and reducing costs. Their proposal uses the modems' CAD capability with an iterative SF inspection and selection algorithm that allows links to operate independently at different data rates, achieving almost 100 % correct detection. This idea had already been implemented on single-channel gateways but had not previously been adapted for multi-hop usage. The authors experimentally evaluate the proposal with up to 10 nodes and compare three topologies (star, tree, and mesh) using Semtech's single-channel SX1272 and multi-channel SX1301 transceivers. Using ASFS allows nodes to choose different and faster SFs, achieving data rates four to six times faster than without it (when all the nodes stick to a common, network-wide slower SF).

Both references in this section are essential to our proposal. They provide the foundations to build faster and more complex LoRa mesh networks with single-channel transceivers, e.g., by choosing the SFs on a per-link basis to maximize the throughput instead of using a [much] slower, common network-wide SF.

While our proposal involves SF detection using preamble detection, similar to the approach presented in [31] and [32], alternative techniques exist to achieve the same goal. Koch et al. introduced a method based on Discrete Wavelet Transform for SF detection that is agnostic to transmitter settings, as detailed in their work [33]. This computationally lightweight algorithm can be implemented on readily available Software-Defined Radios (SDRs), thereby reducing the cost and complexity of multi-SF reception in single-channel implementations.

IV. A MINIMALISTIC DISTANCE-VECTOR ROUTING PROTOCOL FOR LoRa MESH NETWORKS

To assess multi-hop LoRa mesh networks and overcome most of the related work limitations, we propose a proactive, hybrid Layer 2/3 (L2/3) DV RP that takes advantage of this radio technology's specific characteristics. Our design principle is to keep complexity to the minimum needed so that it could be run on resource-constraint embedded devices featuring a microcontroller and a LoRa radio. Besides that, the RP can benefit from LoRa's SFs different range and orthogonality properties, which allow for concurrent transmissions between other pairs of nodes. To achieve it, we also introduce a novel multi-SF-aware ToA metric that minimizes the total transmission time for a packet to reach the destination, which improves performance under certain network conditions.

Our protocol's main features are:

- Distance-vector: best routes towards any destination are calculated in a distributed way across nodes based on the information provided by the neighbors.
- Pro-active: network nodes periodically broadcast available routes, independent of data traffic, refreshing and keeping routes up-to-date and readily available.
- Layer 2+3 hybrid: by aggregating the two layers into one, we simplify and optimize the architecture and the requirements for low-power embedded LoRa devices.
- Duty cycle-aware: for networks operating in unlicensed ISM bands, time-on-air limitations in the form of duty cycles are usually enforced; these restrictions can be embedded into the route metrics calculations.
- Lightweight, flexible, and configurable: many aspects of the protocol (metric, packet timing, etc.) can be fine-tuned to fit specific use cases.
- Concurrent, overlaid networks: thanks to the orthogonality properties of LoRa's SFs, several "virtual" layered networks can operate on the same radio channel, providing higher global throughput.
- ToA metric: route costs are calculated based on the end-to-end packet transmission time, taking multi-SF capability into account.

A. NETWORK TOPOLOGY AND ADDRESSING

The RP builds a flat mesh network topology with no hierarchical differentiation between nodes, regardless of their hardware characteristics or role at the application level. To join the network, a node runs an instance of the RP, generates a local routing table and exchanges routes with its neighboring nodes periodically.

Figure 3 shows the topology of a sample LoRa mesh network running the RP. One-hop, direct communication between a given pair of nodes can usually be performed with different SFs; the diagram indicates the fastest (i.e., the lowest) one that makes the link possible. Communication between distant nodes that are not directly connected is made by multi-hop packet forwarding, using the routes calculated by the RP. While most of the links in the diagram are symmetric, a few require different SFs in each direction to achieve a successful communication. This asymmetry can happen in scenarios with heterogeneous hardware or environmental conditions, either temporary or permanent. As a result, packets traveling between a given pair of nodes could use different routes in each direction.

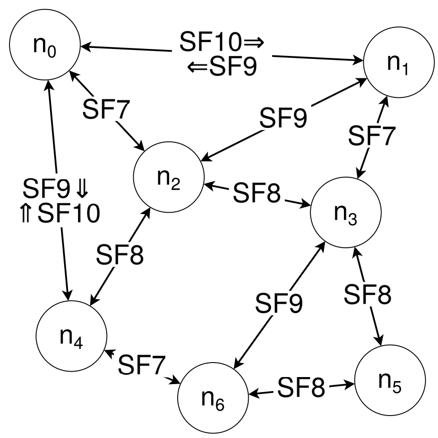


FIGURE 3. Topology of a sample LoRa mesh network. Links between each pair of nodes may use any valid SF (the diagram shows the smallest and fastest possible to create a working link). Most links are symmetrical, but others require different SFs in each direction; this may generate different routes for each direction between a pair of nodes.

To simplify the design of the RP and the applications built upon, we merge the data link (L2) and the network (L3) layers into a single one (compared to, e.g., the Wi-Fi 802.11+IP stack and Ethernet 802.3, where addresses of different types are used on each layer). This solution reduces network traffic overhead and computing effort on the nodes, although it may limit direct interoperation with other networks (e.g., the Internet). In our design, network addresses take 2 bytes, ranging from 0×0000 to 0xFFFF resulting in up to 65.536 usable addresses per network. We consider this specific addressing space size a convenient value between the ability to build an extensive mesh network, LoRa's throughput and range performance, and ease of implementation. However, it could be reduced or increased when implemented in particular scenarios. For example, a smart metering network covering a city with hundreds of thousands of nodes may require a larger addressing space. Still, other strategies (e.g., network partitioning) could be more effective in scenarios.

The RP supports transmitting network packets in unicast to a single node using its unique network address or broadcasting to all the neighbors at one hop using a broadcast address. For the latter mode, the highest address (0xFFFF) is only reserved for broadcast routing packets. Reception acknowledgments, packet retransmissions, etc., can be implemented in higher layers. The RP does not handle them to keep it as simple as possible and reduce its footprint on embedded nodes. Therefore, neither one-hop nor end-to-end delivery is guaranteed at the routing level.

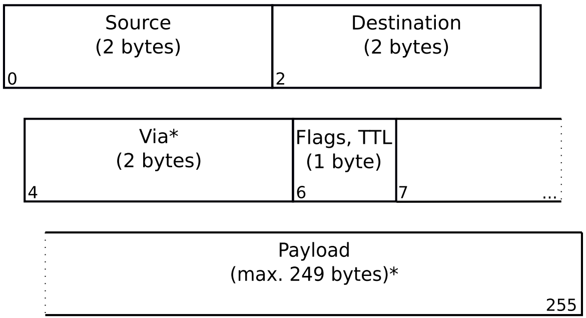


FIGURE 4. Structure of a data/routing packet. ***Source*** and ***Destination*** bytes identify the communication end points. The ***Via*** field is omitted when the broadcast address (0xFFFF) is set as the destination. The ***Flags/TTL*** byte is used to identify the packet type and watch its lifespan. The ***Payload*** field can carry either the RP information or the application data, depending on the packet type.

B. NETWORK PACKETS STRUCTURE

Data and routing packets have a very similar structure, schematically depicted in Figure 4. Since packet size is constrained by the LoRa hardware to a maximum of 256 B, our protocol uses a minimalistic approach to reduce the overhead introduced by the header, which takes 7 B at most.

The first 4 B of a data packet holds the address of the node originating the message (*Source*, 2 B) and the address of the final recipient (*Destination*, 2 B). The following field (*Via*, 2 B) is used in multi-hop data packets to indicate the address of the next hop in its route. Its value changes from hop to hop as intermediate nodes forward the packet. In the last hop in the route, or for single-hop data packets, the *Via* field should be the same as the *Destination*. The *Flags, Time to live (TTL)* space (1 B) is reserved for tagging packets (2 b) if required by the upper application layer and to account for the TTL (6 b) This approach effectively sets a maximum number of hops to 64 for any packet. Last, the *Payload* field holds the actual application data or the RP exchanged between different nodes (see next section), holding up to 249 B.

Routing packets are differentiated from regular data packets because they use the broadcast address in the *Destination* field instead of a unicast address. Therefore, the *Via* field is not necessary and is hence omitted to save 2 B that can be assigned to the payload, which becomes up to 251 B long.

C. ROUTING TABLE

Each node in the network runs an instance of the routing protocol, creating a local routing table that is constantly updated as messages from neighboring nodes are received. Being a DV protocol, the table consists of a list of all the nodes known to be in the network, the neighbor through which they can be reached, the path cost, and the route expiry time. With the information in the table, every node can theoretically communicate with any other node in the network directly (if they are neighbors) or indirectly using multi-hop. Furthermore, every node can also forward multi-hop traffic that goes through it towards its destination.

TABLE 2. Routing table for node n_0 from Figure 3. The first block contains the routes from n_0 to all other nodes. The second block corresponds to routes from neighbor nodes to n_0 (which is required to account for asymmetric links correctly).

Dest	NextHop	SF	Cost	Expiry
0x0001	0x0001	10	-	189
0x0002	0x0002	7	-	293
0x0004	0x0004	9	-	251
0x0001	0x0002	-	4	293
0x0003	0x0002	-	3	293
0x0004	0x0002	-	3	293
0x0005	0x0002	-	5	293
0x0006	0x0002	-	4	293
0x0002	0x0001	-	4	189
0x0003	0x0001	-	12	189
0x0004	0x0001	-	9	189
...	...	-	...	...
Dest	**Source**	**SF**	**Cost**	**Expiry**
0x0000	0x0001	9	-	189
0x0000	0x0002	7	-	293
0x0000	0x0004	10	-	251

In addition to the routes towards other nodes, each node also keeps track of the routes from neighbor nodes to it. Since the physical connection between two neighbor nodes may not be symmetrical, the RP also needs to feed back neighbor nodes with the information of its inbound links so that they can correctly calculate the routes towards it.

Table 2 shows a snapshot of the routing table for node n_0 from Figure 3. In the first block, the three entries on top show the direct, single-hop routes from node n_0 to its neighbors n_1, n_2 and n_4 (indicated by *Dest=NextHop*, but using a different SF). The following entries show multi-hop routes to other routes in the network (indicated by *Dest≠NextHop*). The routes include the nodes that are not directly reachable using one hop (n_3, n_5 and n_6) and also those that have a better metric using a multi-hop path rather than the direct link (this is the case for n_1 and n_3). For instance, the direct single-hop path $n_0 \Rightarrow n_4$ has a higher ToA cost than the multi-hop path $n_0 \Rightarrow n_2 \Rightarrow n_4$.

The second block of entries in Table 2 shows inbound routes to node n_0. Node n_0 does not use them for its own routing decisions. However, it must keep track of them and let its neighbor nodes learn about them. This way, neighbors can know the minimum SF required to reach n_0 and correctly calculate their routes toward it.

Upon successfully transmitting a new routing packet, the receiving node processes it and refreshes its local routing table, updating the information accordingly. First, the link characteristics with the source neighbor node are added to the routing table or refreshed if already present. Then, one by one, the announced routes are processed. Depending on the case, they will be added to the routing table, used to update or replace known routes, or discarded if not useful. The diagram in Figure 5 shows this process as a flow chart.

As exemplified by Table 2, a routing table may contain different entries with routes to the same node via different next hops. While the RP will use the one with the lowest cost by default, this provides alternatives in case the main route

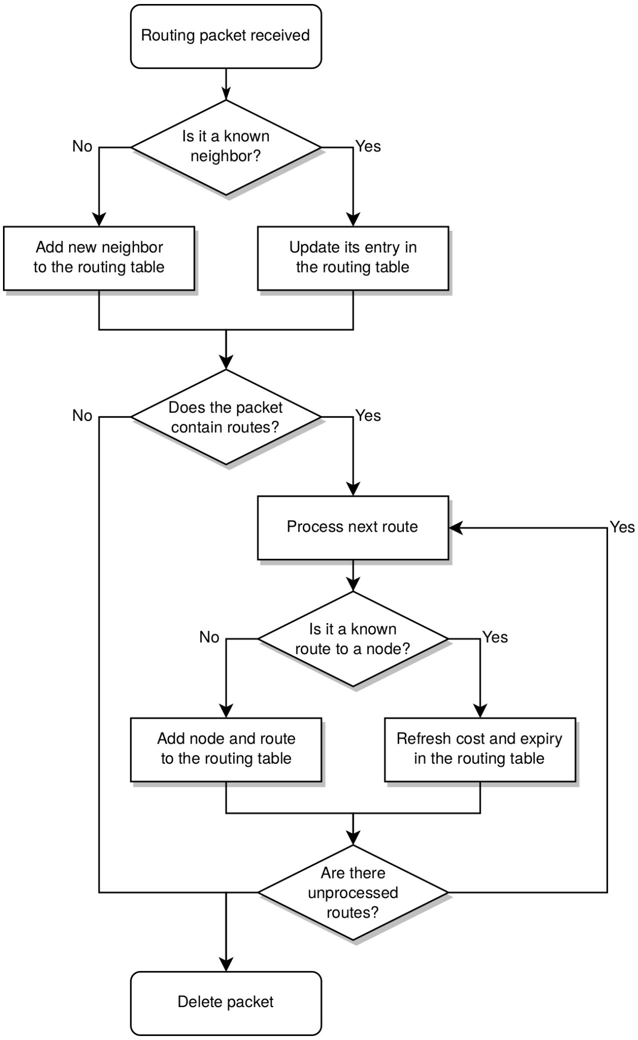


FIGURE 5. Flow chart upon reception of a new routing packet at a node.

suddenly increases its cost or expires. However, maximum total and per-node routes can be established to limit the table's growth.

D. ROUTES EXCHANGE BETWEEN NEIGHBOR NODES

Nodes running an instance of the RP periodically broadcast routing packets to their neighbors. This way, they proactively generate the network topology and keep it updated by refreshing their routing tables locally as packets from neighbor nodes are received. These packets include:

- the source node information (i.e., its address)
- a 6 b incremental counter (instead of the TTL, which can be used to evaluate packet loss and collisions on a link to infer its quality or its occupation)
- an excerpt of the source node routing table (i.e., a list with the best routes and their path cost to the other nodes in the network).

With this information and the details from the LoRa radio physical layer (namely, the SF used to transmit the message), the nodes that receive the broadcast packet update their local routing tables accordingly. In turn, these nodes will propagate their routes further away to their neighbors when they send their own broadcast messages.

The devices running the RP are expected to have single-channel LoRa radios and use the technique described in Section III-C to enable multi-SF reception. Therefore, they can send and receive packets on any SF available between SF_{min} and SF_{max}. The RP can use single-SF metrics, where all the nodes are expected to use a single, network-wide SF, and multi-SF metrics, where nodes may dynamically switch between SFs depending on their current status.

The transmission time for a packet when using a given SF is approximately double the time required by the immediately lower SF.[3] Therefore, when using multi-SF metrics, routing packets are sent using the following strategy. Broadcast messages on any given SF are sent twice as often as on the immediately higher SF (i.e., broadcast packets using SF9 are sent at double the rate than packets using SF10). This method of proceeding balances the cost of using different SFs and helps keep routes using faster links more up-to-date than those using slower links. Broadcast packets on different SFs are sent in random order (rather than sequentially) using the following probability formula:

$$\begin{aligned} p(\mathrm{SF}_n) &= \frac{2^{\mathrm{SF}_{max}-\mathrm{SF}_n}}{\sum_{i=min}^{max} 2^{\mathrm{SF}_i-\mathrm{SF}_{min}}} \\ \text{since:} & \\ \sum_{i=min}^{max} p(\mathrm{SF}_i) &= 1 \\ p(\mathrm{SF}_{min}) &= 2^{\mathrm{SF}_{max}-\mathrm{SF}_{min}} \cdot p(\mathrm{SF}_{max}) \end{aligned} \tag{1}$$

Sending packets on different SFs randomly, instead of using a predefined sequence, also helps to avoid repeated collisions if nodes become synchronized.

The following pseudo-code implements the random selection of a SF between SF_{min} and SF_{max} with the probability described above:

Algorithm 1 Random Selection of the SF on Which to Broadcast a Routing Packet

```
SF = 0;
while SF == 0 do
    thisSF = SFmin;
    while thisSF ≤ SFmax do
        if random(0, 1) < 0.5 ) then
            SF = thisSF;
            break;
        else
            thisSF++;
        end
    end
end
```

[3] To be precise, the LoRa symbol rate is linear with the SF, but the time on air calculation also depends on other non-linear factors. The exact formula can be found in the documentation provided by the chip manufacturer (e.g., Semtech SX1276's datasheet).

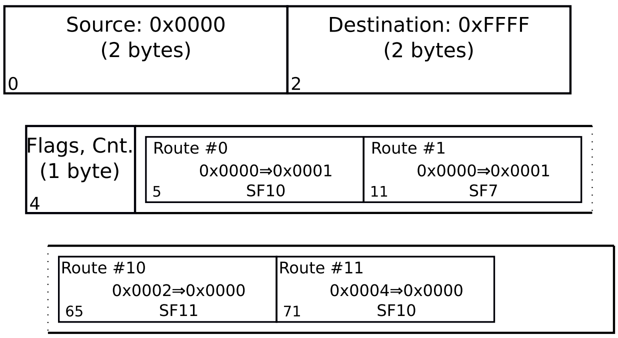


FIGURE 6. **Sample routing packet as sent by node n_0 from Figure 3. The *Destination* address field indicates a broadcast packet destined to its neighbors. Routes in the payload correspond to those in the routing table from Table 2.**

As mentioned in Section IV-B, routing packets are very similar to data packets, except that they are single-hop and use the broadcast address as the destination, skipping the *Via* field. Their payload contains the network routes, as advertised by the source node. Figure 6 shows an example of a routing packet (corresponding to node n_0 from Fig. 3), where the payload contains the best routes of the routing table from Table 2. If the routing table does not fit inside the packet's payload, routes to transmit are chosen randomly by assigning each route a transmission probability as a function of its *Cost* value. This design decision makes routes to nearby nodes more updated than those to distant devices. Therefore, route timeouts must be configured based on the number of nodes and the topology of the specific deployment to avoid flapping.

Upon reception of a new routing packet, nodes process it and refresh their local routing table, updating it accordingly. First, the link characteristics with the source neighbor node are added or refreshed in the routing table. Then, one by one, the received routes (if any)[4] are processed and added or updated in the routing table. The diagram in Figure 5 shows this process as a flow chart.

E. ROUTING METRICS

Our RP can be configured to use different routing strategies, particularly regarding the metric used to calculate the costs of the possible paths towards a destination. These determine which node a packet will be forwarded to reach its destination. The protocol can use a Broadcast (BC) flooding strategy, single-SF metrics (like the well-known Hop Count (HC) or Expected Transmission Count (ETX)), or our proposed ToA multi-SF metric.

1) BROADCAST PACKET FLOODING

The most straightforward multi-hop operation for the RP consists of a packet *flooding* strategy, with no actual routing involved. Nodes keep a copy of the last packets received and broadcast in a finite buffer to avoid double broadcasts that would cause traffic amplification.

A *smarter* version of the flooding strategy, still not involving actual routing, keeps track of neighboring nodes as they send data packets and uses unicast transmissions only for the last hop.

[4] A node joining the network would start with an empty routing table. For a period of time, it would not include any routes in its routing packets but only send to announce itself.

TABLE 3. **Maximum reach, in the simulator, with different SFs.**

SF	Rx sens.	P_{Tx}	BW	Dist.	Δdist.
7	−124 dBm	20 dBm	125 kHz	250 m	*n/a*
8	−127 dBm	20 dBm	125 kHz	349 m	×1, 4
9	−130 dBm	20 dBm	125 kHz	487 m	×1, 4
10	−133 dBm	20 dBm	125 kHz	679 m	×1, 4
11	−135 dBm	20 dBm	125 kHz	848 m	×1, 25
12	−137 dBm	20 dBm	125 kHz	1058 m	×1, 25

2) SINGLE-SF METRICS

Our RP implements four metrics for LoRa mesh networks where nodes use a single network-wide SF. The first and simplest one is HC, for which the cost of a given path equals the number of hops required to reach the destination.

The second metric is ETX, for which the cost of a given path depends on the links along the route, proportional to the number of transmissions required to reach the destination. The quality of a link between two nodes is calculated based on the number of lost routing packets over a period of time (the more lost packets, the worse the link is and the higher its cost).

Two more single-SF metrics are available, which calculate path costs based on the physical links' RSSI. The *RSSI sum* metric uses the reception quality measure to calculate a route cost as a sum of RSSIs along the path:

$$\sum\nolimits_{h=0}^{n} |min(\mathrm{RSSI}_h,\ -1)| \tag{2}$$

so that links with lower signal reception quality incur in a higher route cost. Likewise, the *RSSI product* metric uses the reception quality measure to calculate a route cost as the product of RSSIs along the path:

$$\prod\nolimits_{h=0}^{n} |min(\mathrm{RSSI}_h,\ -1)| \tag{3}$$

3) MULTI-SF TIME ON AIR (TOA) METRIC

The SF is a critical element of the LoRa radio technology, as it poses a trade-off between the transmission reach and the time required to send a packet. Roughly, switching to a SF one step higher (e.g., SF7→SF8) doubles the transmission time (or halves the transmission speed, see Fig. 9), while increasing the distance between ×1, 25 and ×1, 4 (see Table 3).

Taking the restrictions mentioned above into account, we propose a Time on Air (ToA) routing metric to evaluate the cost of a path towards a destination that aims at minimizing the total end-to-end time required to transmit a packet from source to destination and the radio channel occupation.

The ToA metric in our RP calculates the path cost towards a node as a function of the SFs used by the successive links between forwarding nodes until reaching the destination.

It ponders the cost of the hop $h_{l,m}$ between two neighbor nodes n_l and n_m as a power of 2:

$$\mathrm{h}_{l,m} = 2^{\mathrm{SF}_{l,m}-\mathrm{SF}_{min}} \tag{4}$$

where $\mathrm{SF}_{l,m}$ is the smallest SF required to successfully transmit between nodes n_l and n_m, and SF_{min} is the minimum SF available in the system.[5] Therefore, we calculate the cost of a path between two arbitrary nodes n_i and n_j, with H intermediate hops, as:

$$\mathrm{ToA}_{i,j} = \sum_{k=1}^{H} \mathrm{h}_k = \sum_{k=1}^{H} 2^{\mathrm{SF}_k-\mathrm{SF}_{min}} \tag{5}$$

where SF_k corresponds to the SF used in each of the intermediate H hops in the route. If two or more paths are available with the same metric, the one with the next hop using a lower SF is preferred. Still, in case of a tie (same metric and same SF in the next hop), the path is chosen randomly among the contenders.

The example in Figure 3 uses SF7 as the smallest SF available (SF_{min}). Using the ToA metric, the cost of the direct single-hop path from node n_0 to node n_1 would be $2^{10-7} = 8$. Instead, the three-hops path via nodes n_2 and n_3 would be preferred, since its ToA metric would be $2^{7-7} + 2^{8-7} + 2^{7-7} = 1 + 2 + 1 = 4$. The usage of a simple additive metric calculation is convenient in the context of resource-constraint IoT devices, which are often driven by 8 or 32 bit Microcontroller Units (MCUs). These operations can be easily implemented and require few processor cycles.

Section VI evaluates the metric in depth in a simulation environment and compares it with other well-known metrics.

F. LOOPS MITIGATION

Our RP is subject to routing loops (e.g., in the case of a node disappearing from the network), which is a common problem in DV algorithms. To mitigate them or their effects, we implement the following mechanisms.

1) ROUTE POISONING

If a direct route (i.e., to a one-hop neighbor) reaches the timeout on a given node, it broadcasts this route with the maximum possible cost. This way, nodes will drop the expired route sequentially and propagate the action along the network.

Given that routing packets are transmitted without reception guarantees, this mechanism may have a limited impact on loop mitigation, especially on busy networks with high packet collision probability. As a possible workaround, *poisoned* routes could be broadcast twice or more before being deleted locally.

2) HOLD-DOWN

When a node learns about an unreachable route (e.g., a failed neighbor), it sets up a timer. During this period, the node ignores incoming information about the route, avoiding an infinite count situation.

This mechanism may be only partially effective, depending on whether a node can detect or receive a route's unreachability before starting an infinite count. Additionally, it may increase network convergence times.

3) MAXIMUM ROUTE COST

Routes have a finite maximum cost, and those reaching this value are discarded. Therefore, in the event of an infinite count loop, affected routes are eventually discarded.

This parameter poses a trade-off between how fast loops are fixed and how large a network can be regarding routing (i.e., maximum number of hops).

4) TTL

Data packets have a maximum hop count of 64 hops. However, this limit can be set to a lower value, like the number of known nodes in the network (if smaller than 64). This way, packets entering a routing loop will be discarded sooner, as their TTL is exhausted.

5) AVOID DUPLICATE TRANSMISSIONS

When the flooding mechanism is used (Section IV-E1) instead of actual routing, nodes hold a copy of the last packets they forwarded in a finite buffer. If the packet is received again to be forwarded, it is discarded, avoiding duplicates and loops. This mechanism is also employed when actual routing is implemented and complements the TTL embedded in the packets to avoid them entering routing loops.

G. PROTOCOL CONFIGURATION

Many parameters of the RP can be configured to adapt to specific use cases and conditions. This section summarizes the most remarkable ones for the RP operation.

First, several LoRa-related aspects can be customized. For instance, the maximum SF to use can be changed (from the default SF12 to a smaller one, which could be required in certain regulatory domains). Instead of the default ToA metric, other well-known ones can be chosen (HC, ETX, etc.). Other aspects, like the expiry time of a learned route, can be modified to ensure the liveliness of the information in the routing table. Table 4 shows these configurable parameters. Regarding data and routing packets, their transmission priority at a node can be adjusted in an $n : 1$ ratio. Similarly, forwarded and local data packets can be prioritized according to the application's needs.

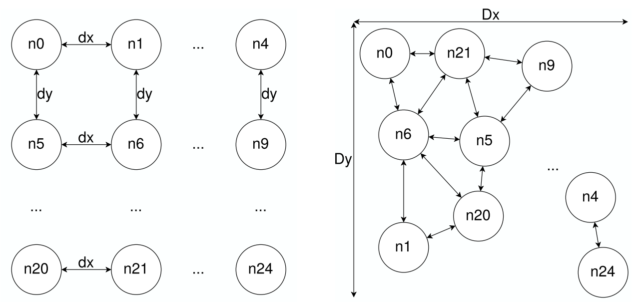
V. EVALUATION METHODOLOGY

In this study, we use OMNeT++ [8], an extensible, modular, component-based C++ simulation library and framework, in combination with FLoRa [7], a framework to carry out end-to-end simulations for LoRa networks. OMNeT++ is a well-known discrete event simulator framework used by a lively academic community and, together with FLoRa, it provides an implementation of the LoRaWAN architecture [34]

[5] The smallest SF available in LoRa is SF6. For practical reasons, the minimum SF commonly used is SF7, but a higher value could be preferred.

TABLE 4. Configuration options for the RP.

Scope	Parameter	Range	Default
LoRa	Minimum SF	SF7, ... SF_{max}	SF7
LoRa	Maximum SF	SF_{min}, ... SF12	SF12
Routing	Metric	BC [a], HC, ETX, RSSI (sum/product), ToA	ToA
Routing	Average routes broadcast period	$0 \dots \infty$	60 s
Routing	Routes expiry time (SF_{min})	$0 \dots \infty$	300 s
Routing	Max. routes to a node	$1 \dots n$	2
Routing	Max. total routes	$0 \dots 1024$	1024
Routing	Routing/Data traffic prior	$n : 1$	$10 : 1$
Routing	Forward/Local traffic prior	$n : 1$	$10 : 1$
Regulation	Duty cycle (%)	$(0, 100) – 100$	100



(a) Symmetric grid topology, with equal horizontal (d_x) and vertical (d_y) spacing between nodes.

(b) Random topology, with nodes distributed over a $D_x \times D_y$ area using a uniform density probability function.

FIGURE 7. Depiction of the network topologies used in the simulations.

and an accurate model of the LoRa radio physical layer derived from previous experimental findings [35]. Its features and performance have been analyzed, evaluated, and compared to other available tools [36].

We have stripped down the LoRaWAN functionalities from the FLoRa framework, allowing direct communication and packet forwarding between end nodes without a gateway. Our solution includes adding a DV routing protocol with different path cost calculation metrics. Furthermore, we have added to the framework other helpful features, like the CAD found in other implementations [31], [32]. The source code for our derived project, rebranded as FLoRaMesh, is publicly available on GitLab [37].

A. NETWORK TOPOLOGIES

The simulations consist of a network with a variable number of nodes arranged in two different topologies: an $N \times N$ symmetrical grid topology with equal vertical and horizontal distance (Fig. 7a) and a random topology with N^2 nodes uniformly distributed over a delimited square area (Fig. 7b).

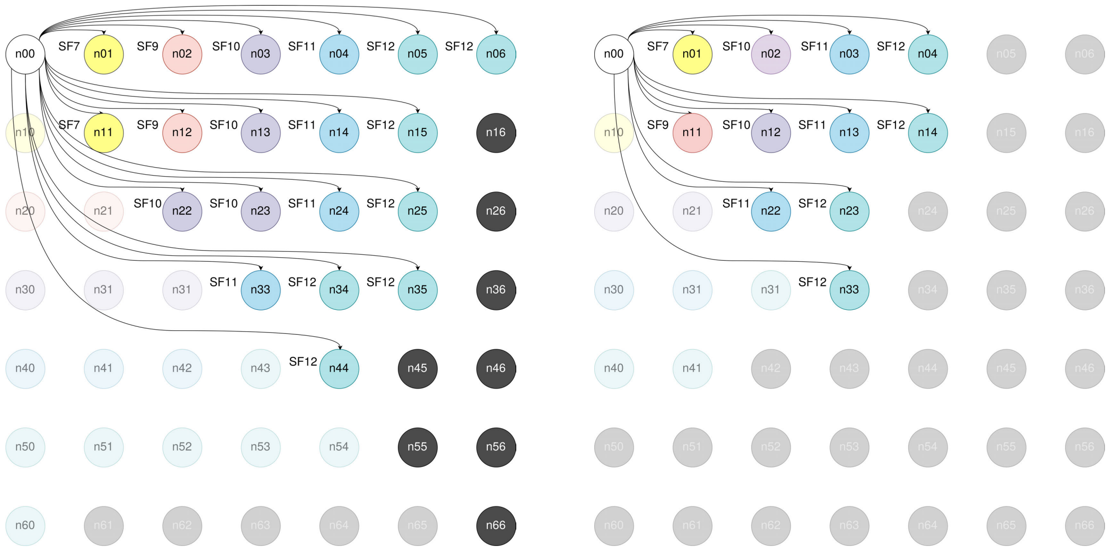
Each of these two topologies serves a different evaluation purpose. On the one hand, the grid topology with a constant distance between nodes offers a regular and predictable environment. Therefore, once the right SF and transmission power are set, single SF routing provides communication to either all or none of the nodes, and performance will only depend on the metric properties (rather than on the network topology characteristics). This procedure also allows checking if multi-SF routing can offer any advantage (i.e., whether using one given SF and, concurrently, in addition, higher ones, has any impact on performance, compared to using just one). On the other hand, the random topology allows measuring how multi-SF routing adapts to heterogeneous situations and how using different SFs in different parts of the network (i.e., use the fastest SF possible for each link between pairs of nodes) performs compared to using network-wide common SF that provides communication to all the participating nodes.

When nodes are arranged following a grid topology, four different spacing between them are used (both vertical and horizontal): 177 m, 178 m, 246 m and 247 m. These values are not arbitrarily chosen but have a specific purpose, as depicted in Figure 8. A spacing between of 177 m allows nodes using the shortest-range SF7 to communicate with their adjacent nodes in horizontal, vertical, and diagonal (Fig. 8a). When the spacing is increased by one unit, diagonal communication with adjacent nodes is no longer possible with SF7, only vertically and horizontally. Similarly, a 246 m spacing or 247 m allows communication between adjacent nodes in diagonal with SF8 or requires using the slower SF9. The experiments with these four spacing values allow a comparison of how the mesh density affects the network performance.

When the random topology is used, nodes are uniformly distributed over the same area occupied by the grid topology. For instance, in a network with $N^2 = 36$ nodes, to compare it with a grid topology where nodes are spaced 178 m: $5 \cdot 178\,\text{m} \times 5 \cdot 178\,\text{m} = 890\,\text{m} \times 890\,\text{m}$). This approach allows comparison between a synthetic network and a more heterogeneous deployment.

B. NODES CHARACTERISTICS

All the nodes are identically configured in the simulation, using the same settings for the LoRa physical layer (e.g., transmission power, bandwidth, etc.), as listed in Table 5. Therefore, their behavior and performance are only affected by their network position and interaction with neighbors are only affected by their behavior and performance. Most of the chosen configuration parameters (SF, bandwidth, preamble size, etc.) are typical in real-world deployments [38]. Given these settings in the simulator, two nodes using SF7 can



(a) $d_x = d_y = 177\,\mathrm{m}$ (b) $d_x = d_y = 247\,\mathrm{m}$

FIGURE 8. Minimum SFs required for a LoRa packet to successfully reach different nodes from node n_{00}.

TABLE 5. Common node settings in the simulations.

Configurable parameter	Values
Carrier frequency	868 MHz
Channel bandwidth	125 kHz
Spreading Factor (SF)	7, 8, 9, 10, 11, 12 ^{a}
Transmission power	20 dBm
Coding rate (CR)	4/5
Preamble length	16 symbols

successfully communicate up to 250 m apart. Therefore, to reach nodes further away, a node either must switch to higher SFs, use packet forwarding via its neighbors, or combine both strategies. This maximum communication distance also motivates the spacing described previously in Section V-A.

For completeness, Table 3 summarizes the maximum transmission distance a node can reach in the simulator, for all the available SFs. The values depend on the LoRa settings specified in Table 5 and on the radio propagation model implemented in the simulator framework (in this case, an urban environment). With a maximum of 1058 m, when using SF12, communication between nodes further away is only possible through multi-hop.

C. TRAFFIC SETTINGS

During a simulation run, each node generates a total of $100 \times (N^2 - 1)$ unique data packets (i.e., 100 different packets for each one of the other nodes in the network) with a fixed size of 27 B (header+payload). The packets are progressively sent during the simulation, at regular intervals of 100, 10, 1, and 0.1 s (corresponding to a low, medium, high, and saturation traffic respectively), towards their destinations in random order, thus evenly distributing traffic between all the nodes in the network.

D. PRELIMINARY SIMULATOR AND LoRa HARDWARE BENCHMARK

We run a simple benchmark to verify that the devices simulated with the FLoRa framework offer a network performance comparable to real ones. The experiment consists of two nodes, one transmitting packets continuously for one hour and the other listening to them. Duty cycle restrictions are not enforced, letting the transmitter occupy 100 % of the airtime.

We perform several simulations using different payload sizes with all the SFs, 7 to 12, and count the number of received packets to calculate the achieved throughput. In parallel, we run the same experiment with real hardware, using two TTGO ESP32 devices[6] and the RadioLib library to interact with the LoRa transceiver [39]. Table 6 provides the node configuration and settings used in simulated and real environments. Note that since the real devices are deployed in the lab, only 2 m apart, the lowest transmission power is used, and a 15 dB attenuator is coupled to the antenna to minimize radio emissions to the environment.

Figure 9 shows the achieved throughput from the source to the destination, for both the FLoRa simulation and the physical devices. Values range from 7.4 bps (smallest 1 B payload using SF12) to roughly 5 kbps (250 B payload size or bigger, using SF7). While the simulator and real hardware provide similar results, noticeable deviations occur. In particular, achieved throughput with lower SFs is up to 6 % higher in real hardware than in the simulator. This behavior could be attributed to how the software calculates the packet transmission time. Also, the performance of hardware devices appears slightly lower with large payloads. This decrease in performance could be caused by delays in the

[6] LILYGO® TTGO ESP32: http://www.lilygo.cn/prod_view.aspx?TypeId=50003&Id=1271&FId=t3:50003:3

TABLE 6. Node settings for the benchmark experiment.

Configurable parameter	Values
Carrier frequency	868 MHz
Link bandwidth	125 kHz
Spreading Factor (SF)	7
Transmission power	20 (FLoRa), 2 dBm (TTGO)
Coding rate (CR)	4/5
LoRa packet preamble	16 symbols
Message header size	7 bytes
Message payload size	1 to 500 bytes[a]
Time between packets	0 s
Duty cycle	100 % (i.e., no restriction)

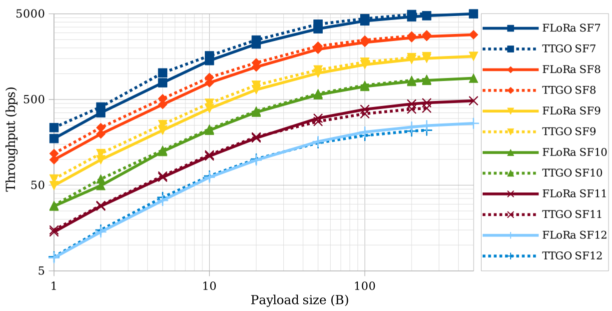


FIGURE 9. Maximum unidirectional data throughput, in bps, with different payload sizes, for SFs 7 to 12. Solid lines correspond to the FLoRa simulator. Dotted lines are obtained from two TTGO ESP32 devices.

embedded System on a Chip (SoC) when receiving packets from the LoRa transceiver and processing them. Considering the magnitude of these discrepancies, we assume that the modifications made to the FLoRa framework do not affect its accuracy so that it provides a simulation environment precise enough.

E. DUTY CYCLE

Radio spectrum regulators usually enforce duty cycle limitations. For example, the unlicensed ISM bands where LoRaWAN is commonly used are subject to a 10 % or a 1 % duty cycle in most parts of the world. However, LoRa radios can also be used in bands without limitations on the transmission time, whether licensed or not (e.g., the 2.4 GHz band).

The FLoRa framework, in its original form, enforces the duty cycle restrictions found in ISM bands to the simulations. In our derived work, FLoRaMesh, we implemented the duty cycle enforcement mechanism at the routing protocol level. A user-configurable parameter in the (0, 1] interval is employed to define the average fraction of time that a node can spend transmitting messages.

VI. EXPERIMENTAL RESULTS

This section evaluates the proposed RP for LoRa mesh networks. We conduct several experiments using different topologies, traffic settings, and node configurations to obtain performance details about three significant Key Performance Indicators (KPIs): *scalability*, *throughput*, and *latency*.

To get a better understanding of the protocol capabilities, we perform the set of experiments using the different multi-hop and routing strategies presented in IV-E: Broadcast (BC), Hop Count (HC), Expected Transmission Count (ETX), Received Signal Strength Indicator (RSSI)[7] and multi-SF ToA.

A. SCALABILITY

The first KPI we analyze is scalability to understand how a LoRa mesh network behaves as the *number of nodes* grows. A network can scale up when adding more nodes, covering a larger area, or keeping it fixed and increasing the density of the devices.

To understand the behavior of our RP with the LoRa radio technology regarding scalability, we perform two sets of experiments for both scenarios mentioned above and analyze the effects of each. First, we simulate a mesh network as it increases its number of nodes, keeping the nodes at a constant distance, covering a larger area each time. Second, we analyze a mesh network covering a predefined fixed area with an increasing number of nodes and, hence, an increasing density.

To evaluate our multi-SF routing protocol based on the ToA metric, we test it under low, medium, and high traffic conditions and compare it with the single-SF BC, HC, ETX and RSSI routing strategies. We simulate network deployments with $N^2 = \{9, 16, 25, 36, 49 \text{ and } 64\}$ nodes, both on a grid topology or randomly distributed over an equivalent area. The ToA metric is tested with all the possible SF ranges (i.e., SF7-8, SF7-9, SF7-10, SF7-11, SF7-12, then SF8-9, SF8-10, SF8-11, etc.).

1) AREA COVERAGE

In this section, we perform different network simulation experiments to investigate network scalability. An increasing number of nodes covers an area that grows proportionally. We look at the end-to-end PDR performance metric of the data packets sent between any pair of nodes in the network. We hypothesize that, as more nodes operate on the network and, on average, the distance between a pair of nodes will be higher, end-to-end transmissions will require more hops, which will result in a higher number of collisions, hence increasing packet loss and reducing PDR. Our objective is to find whether using the multi-SF ToA metric and combining different SFs in the network improves the average PDR or not.

Initially, we simulate network deployments on a regular grid topology, keeping horizontal (d_x) and vertical (d_y) distance between nodes constant. Thus, as the number of nodes increases, so does the area they occupy. By testing different values for these distances $d_x = d_y$, we aim to understand if the routing protocol can take advantage of the multi-SF capabilities of LoRa and improve performance. Later, we perform the same network simulations except that

[7] Simulations with RSSI sum and RSSI product reported almost identical results, so only the first one is shown in the results for simplicity.

TABLE 7. List of topologies, number of nodes, distance between nodes, and the total area used in the simulations.

Topology	Nodes	$d_x = d_y$	Area
Grid	9	177 m	$354 \times 354\,\mathrm{m}^2$
		178 m	$356 \times 356\,\mathrm{m}^2$
		247 m	$494 \times 494\,\mathrm{m}^2$
		248 m	$496 \times 496\,\mathrm{m}^2$
Random (uniform)	9	n/a	$354 \times 354\,\mathrm{m}^2$
		n/a	$356 \times 356\,\mathrm{m}^2$
		n/a	$494 \times 494\,\mathrm{m}^2$
		n/a	$496 \times 496\,\mathrm{m}^2$
Grid	16	177 m	$531 \times 531\,\mathrm{m}^2$
		178 m	$534 \times 534\,\mathrm{m}^2$
		247 m	$831 \times 831\,\mathrm{m}^2$
		248 m	$834 \times 834\,\mathrm{m}^2$
Random (uniform)	16	n/a	$531 \times 531\,\mathrm{m}^2$
		n/a	$534 \times 534\,\mathrm{m}^2$
		n/a	$831 \times 831\,\mathrm{m}^2$
		n/a	$834 \times 834\,\mathrm{m}^2$
...	...	...	...
Grid	64	177 m	$1239 \times 1239\,\mathrm{m}^2$
		178 m	$1246 \times 1246\,\mathrm{m}^2$
		247 m	$1729 \times 1729\,\mathrm{m}^2$
		248 m	$1736 \times 1736\,\mathrm{m}^2$
Random (uniform)	64	n/a	$1239 \times 1239\,\mathrm{m}^2$
		n/a	$1246 \times 1246\,\mathrm{m}^2$
		n/a	$1729 \times 1729\,\mathrm{m}^2$
		n/a	$1736 \times 1736\,\mathrm{m}^2$

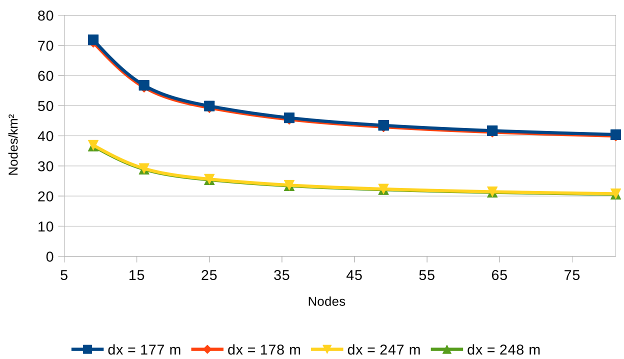


FIGURE 10. Nodes density (nodes/km^2) in function of the number of nodes simulated, for the different spacing between nodes d_x used in Sec VI-A1. The graphs show an asymptotic behavior.

nodes are randomly distributed, occupying an equivalent area to the previous grid topology, creating a more heterogeneous topology. Table 7 lists the topologies and dimensions used in these simulations.

Due to the shape of the grid topology and the border effect, the density of nodes (i.e., the nodes/area ratio) is not constant, as shown in Figure 10. However, as the network grows, it tends asymptotically to a fixed value, only determined by the number of nodes and the spacing between them. From 36 nodes onward, we consider that the border effect is negligible, so the resulting PDR will be affected mainly by the number of nodes and the area they occupy rather than by their density.

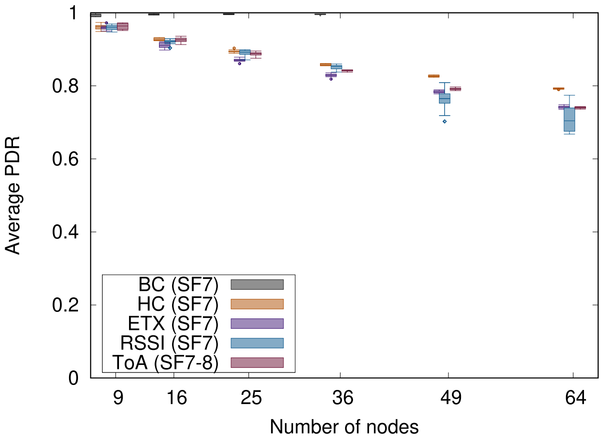
Figure 11 shows the average PDR as a function of the number of nodes deployed with a grid topology when different routing strategies are used. Similarly, Figure 12 corresponds to the same experiments, with nodes randomly deployed over the equivalent areas. Each row in the figure corresponds to a different spacing between nodes (medium spacing omitted, minor difference), and each column corresponds to low- to high-network traffic scenarios (medium traffic omitted, minor difference).

The general trend shows the expected negative relation between the PDR and the number of nodes –except for the flooding-based strategies–. First, we can observe that in low traffic scenarios, BC-based forwarding provides the best PDR results, with respectively 99 % of the packets successfully delivered end-to-end. The flooding strategy's redundancy, combined with the relatively free nodes, allows room for packet duplication, which helps ensure that at least one of the copies reaches the destination. This approach, however, comes at the expense of high latency, as packets are often queued in the intermediate nodes. As the traffic increases and the links begin to saturate, BC provides no significant benefits compared to other routing strategies.

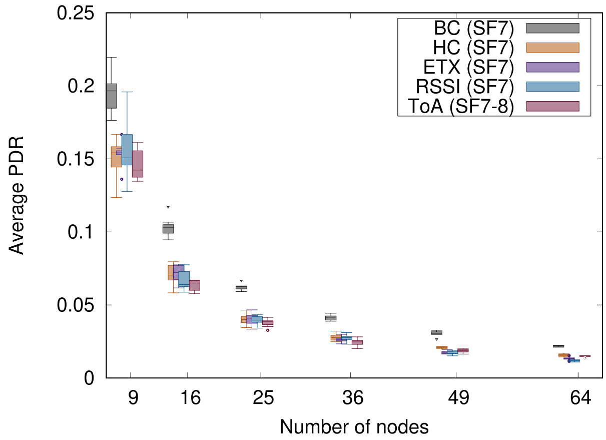
Regarding the actual single-SF routing strategies, HC (the most straightforward metric) provides the best PDR performance in most of the experiments, closely followed by the RSSI metric. The traffic-aware ETX metric performs worse in low-traffic scenarios but is almost on par with HC as the traffic load increases. The multi-SF ToA does not offer any significant improvement in terms of PDR, and its performance is in between the single-SF metrics. Furthermore, it does not take advantage of the available links in diagonal, using a higher SF that would allow using two different SFs simultaneously. Therefore, in the context of a grid topology with equal spacing between all nodes, a simpler single-SF algorithm seems to be the most reliable solution for end-to-end packet delivery. Despite not being shown in the graphs, goodput and latency performance figures are consistent with the PDR results.

The simulations with random network topologies offer a different picture, as visible in Figure 12. There, we can observe that the multi-SF ToA routing metric provides better PDR performance than single-SF metrics in some situations. More specifically, where nodes are more separate apart, for low traffic load (Figs. 12c the ToA result bars have less dispersion, and their averages are higher along most of the experiments. These results indicate that the multi-SF routing strategy has a direct impact on nodes that, because of their random placement, are more isolated (e.g., at the edges of the network) and, therefore, have fewer communication opportunities than those at the center of the network area. In other words, the ToA metric better adapts to heterogeneous network topologies and balances the PDR performance between nodes with different network centrality. These benefits are also visible regarding the goodput KPI, which is analyzed later in Section VI-B.

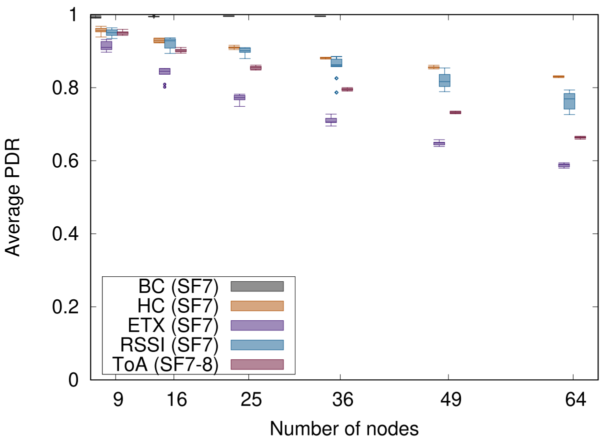
In conclusion, multi-SF routing strategies like our proposed ToA metric may have a positive impact on PDR, which is a main indicator of the scalability, depending on the network topology. They can better cope with the network links' heterogeneity than other routing strategies, benefiting



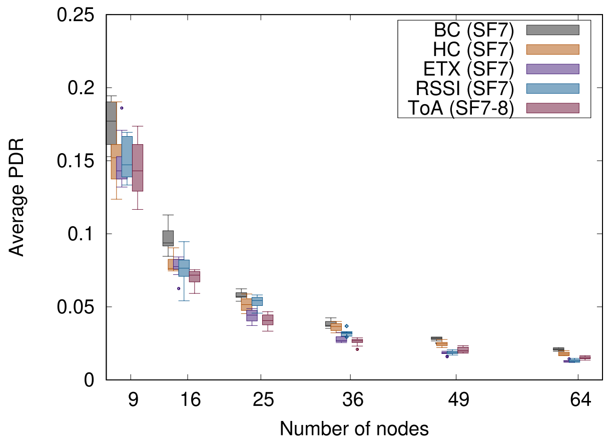
(a) $\mathrm{d}_x = 177\,\mathrm{m}$, low traffic load.



(b) $\mathrm{d}_x = 177\,\mathrm{m}$, high traffic load.



(c) $\mathrm{d}_x = 248\,\mathrm{m}$, low traffic load.



(d) $\mathrm{d}_x = 248\,\mathrm{m}$, high traffic load.

FIGURE 11. Average network PDR for different numbers of nodes in a grid topology with a constant horizontal and vertical node spacing of 177 up to 248 m, using different routing strategies. Notice the *y*-axis scale change.

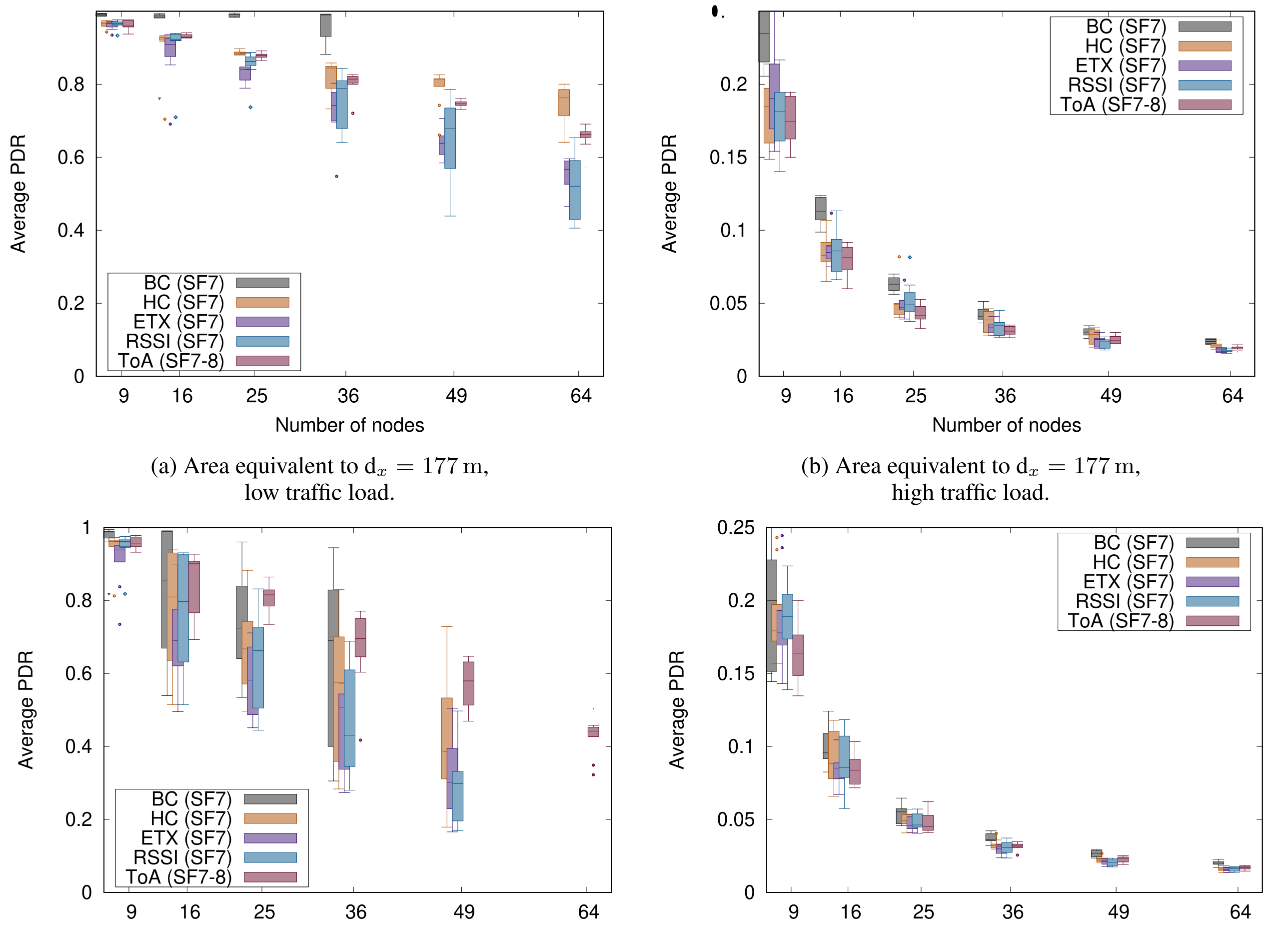
overall performance. Still, they do not provide any advantage to networks with regular topologies like a grid one. Since diversity in nodes and links is expected to occur in real-world systems, with nodes placed in diverse locations, subject to different environmental conditions (attenuation, interference, number of neighbors, etc.), the multi-SF ToA metric may ease the deployment of LoRa mesh networks and improve their performance –at least, in terms of end-to-end PDR–.

2) DENSITY OF NODES

In this second part of the section, we conduct several experiments to analyze scalability as the number of nodes grows and their density increases. We use the end-to-end PDR, averaged among all the nodes in the network, as the performance metric. As more nodes are placed in the same area and radio transmissions become more frequent, the PDR is expected to degrade due to the higher collision probability. We aim to find whether the simultaneous usage of multiple SFs reduces collisions, improving the average PDR.

We simulate different network deployments, with $N^2 = \{9, 16, 25, 36, 49$ and $64\}$ nodes on a fixed area of $500 \times 500\,\mathrm{m}^2$. As the number of nodes grows on each iteration, while the area is kept constant, the density of nodes becomes higher proportionally. To evaluate our multi-SF routing protocol based on the ToA metric, we test it under low, medium, and high traffic conditions and compare it with the BC, HC, ETX and RSSI routing strategies. The ToA metric is tested with all the possible SF ranges (i.e., SF7-8, SF7-9, SF7-10, SF7-11, SF7-12, SF8-9, SF8-10, SF8-11, etc.).

Figures 13 and 14 show the average PDR in function of the number of nodes used in the simulations when different routing strategies are used. In the former, nodes are deployed on a grid topology; in the latter, nodes are randomly deployed with uniform distribution (details about it are in Sec. V-A). Each figure contains two sub-figures corresponding to low and high network traffic conditions (medium omitted, minor difference)). As expected, their general trend shows a negative relation between the average PDR and nodes' number –and density. For the case of the ToA metric, the SFs range providing the best performance results (e.g., SF7-9) is plotted (rather than all the combinations).



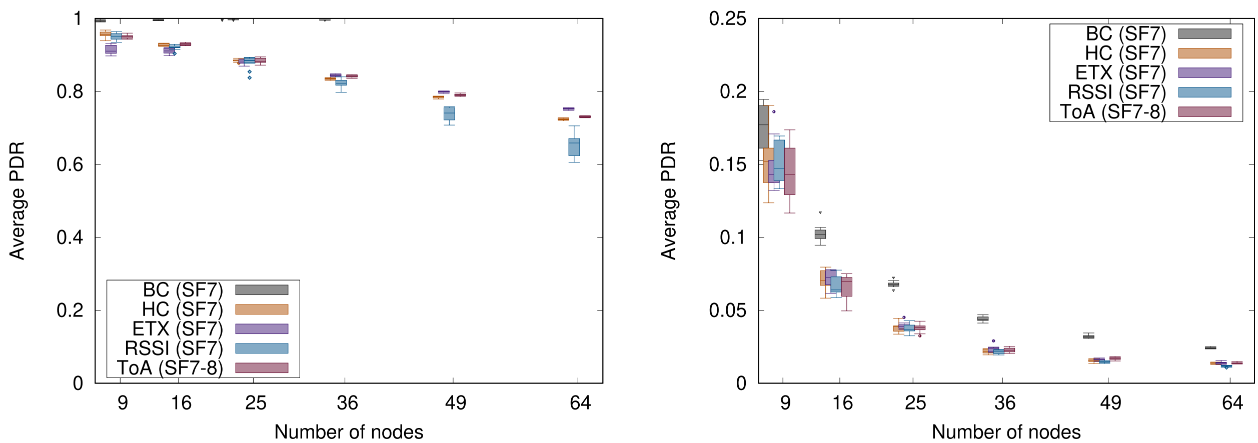
(a) Area equivalent to $d_x = 177\,\mathrm{m}$, low traffic load.

(b) Area equivalent to $d_x = 177\,\mathrm{m}$, high traffic load.

(c) Area equivalent to $d_x = 248\,\mathrm{m}$, low traffic load.

(d) Area equivalent to $d_x = 248\,\mathrm{m}$, high traffic load.

FIGURE 12. Average network PDR for different numbers of nodes in a random topology with a constant horizontal and vertical node spacing of 177 up to 248 m, using different routing strategies. Notice the *y*-axis scale change.

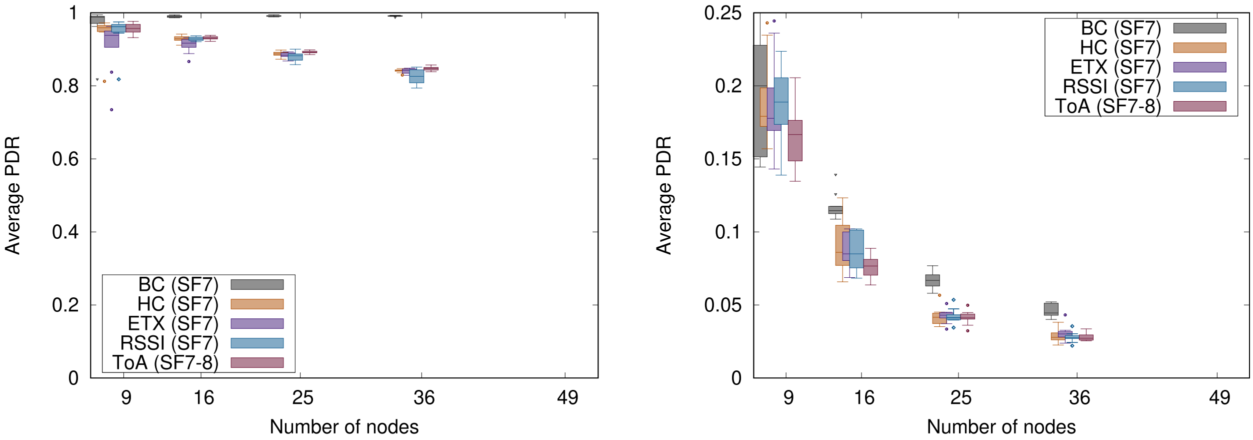


(a) Low traffic load.

(b) High traffic load.

FIGURE 13. Average network PDR, for different number of nodes, in a grid topology on a fixed area of 500 × 500 m^2, using different routing strategies. Notice the *y*-axis scale change.

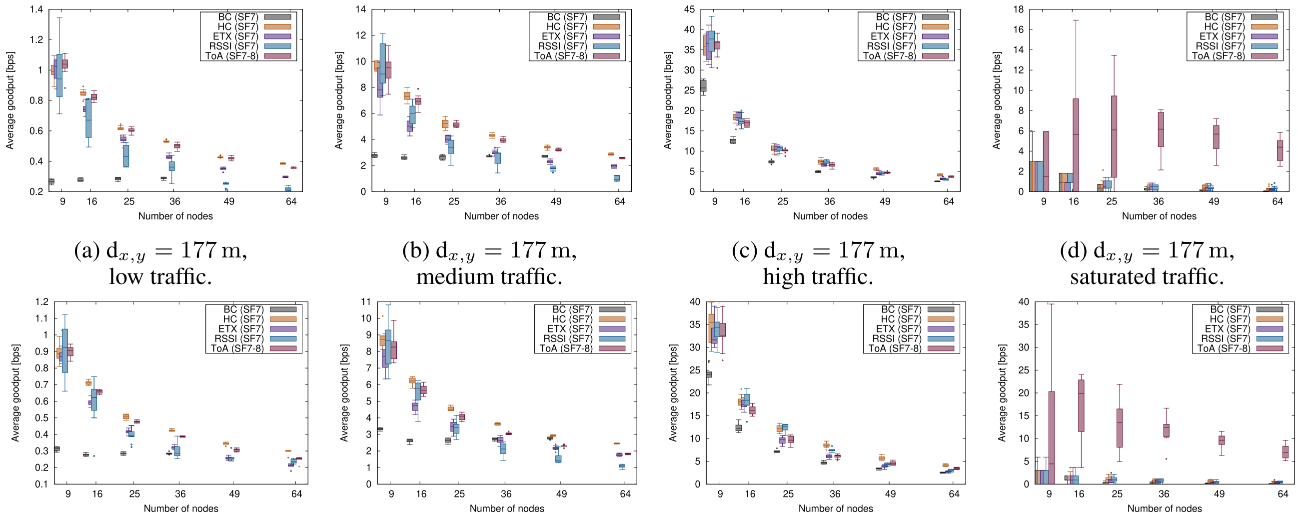
For the low traffic scenarios (Figs. 13a, 14a), the BC strategy provides the best PDR results. Since packets are broadcast and replicated on each hop, flooding the network, the chance for any copies to arrive at the destination is very high, mostly compensating for any additional collisions. The actual routing strategies (HC, ETX, RSSI and ToA) offer



(a) Low traffic load.

(b) High traffic load.

FIGURE 14. **Average network PDR, for different number of nodes, in a random topology with uniform distribution on a fixed area of 500 × 500 m^2, using different routing strategies. Notice the *y*-axis scale change.**



(a) $\mathrm{d}_{x,y} = 177\,\mathrm{m}$, low traffic.

(b) $\mathrm{d}_{x,y} = 177\,\mathrm{m}$, medium traffic.

(c) $\mathrm{d}_{x,y} = 177\,\mathrm{m}$, high traffic.

(d) $\mathrm{d}_{x,y} = 177\,\mathrm{m}$, saturated traffic.

(e) $d_{x,y} = 248\,\mathrm{m}$, low traffic.

(f) $\mathrm{d}_{x,y} = 248\,\mathrm{m}$, medium traffic.

(g) $\mathrm{d}_{x,y} = 248\,\mathrm{m}$, high traffic.

(h) $\mathrm{d}_{x,y} = 248\,\mathrm{m}$, saturated traffic.

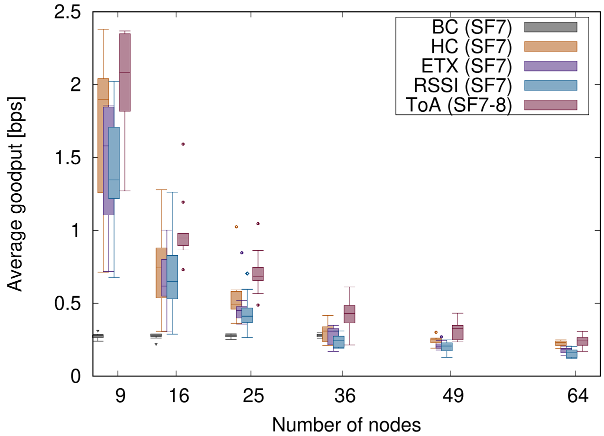
FIGURE 15. **Average network goodput, in bps, for different number of nodes in a grid topology with a constant horizontal and vertical node spacing of 177 and 248 m, using different routing strategies. Notice the *y*-axis scale changes.**

a quasi-linear relation between the number of nodes and the PDR. Their results fluctuate depending on the number of nodes and the topology but are consistent, revealing no evident performance difference between them. As the traffic increases (Figs. 13b, 14b, the network becomes each time more congested, affecting the collision probability and hence PDR. The difference between BC and the routing strategies becomes smaller, as the congested network does not allow room for packets to flood the whole network and reach the destination.

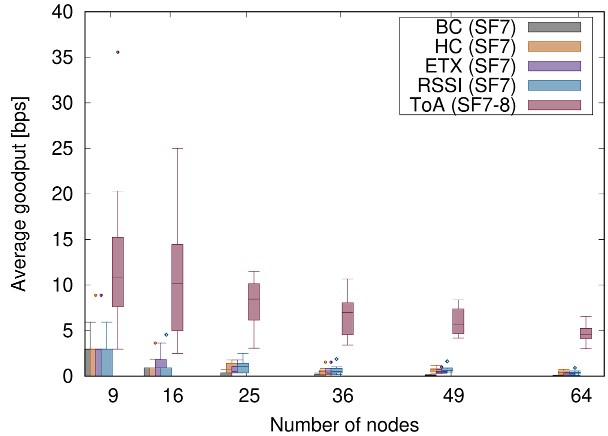
According to the PDR results, there is no observable benefit in using the multi-SF approach and the ToA instead of any other single-SF strategy. Our simulations indicate, however, that combining different SFs does help to increase PDR results in very congested networks, where nodes are constantly sending packets one after the other. In this scenario, –regardless of the topology–, single-SF routing can barely provide a PDR of 0.001 (i.e., one packet out of a thousand reaching its final destination) in a 16-nodes network, while multi-SF ToA achieves, approximately, a PDR of 0.01 (i.e., one packet out of a hundred). Even if the effect of multi-SF is measurable, it is irrelevant in practical terms, so the results were removed from Figures 11 and 12.

B. THROUGHPUT

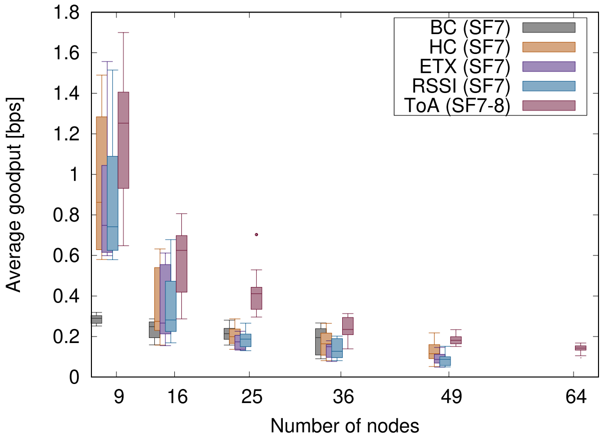
Besides the scalability and the PDR analyzed above, *throughput* is an important KPI to take into account to understand the amount of data a LoRa-based mesh network can handle. In Section V-D, we ran a simple benchmark that indicated the maximum throughput two LoRa nodes could achieve in



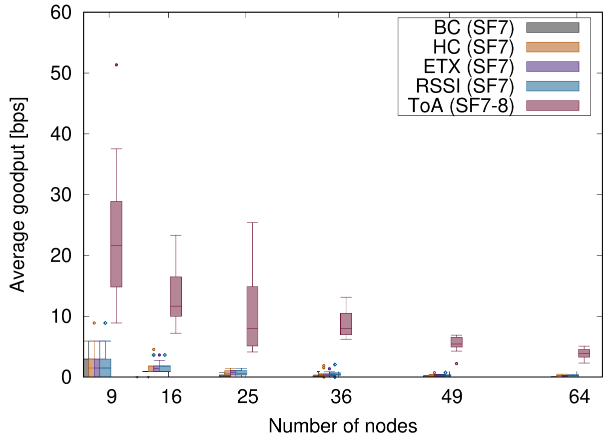
(a) Area equivalent to $d_x = 177\,\text{m}$, low traffic.



(b) Area equivalent to $d_x = 177\,\text{m}$, saturated traffic.



(c) Area equivalent to $d_x = 248\,\text{m}$, low traffic.

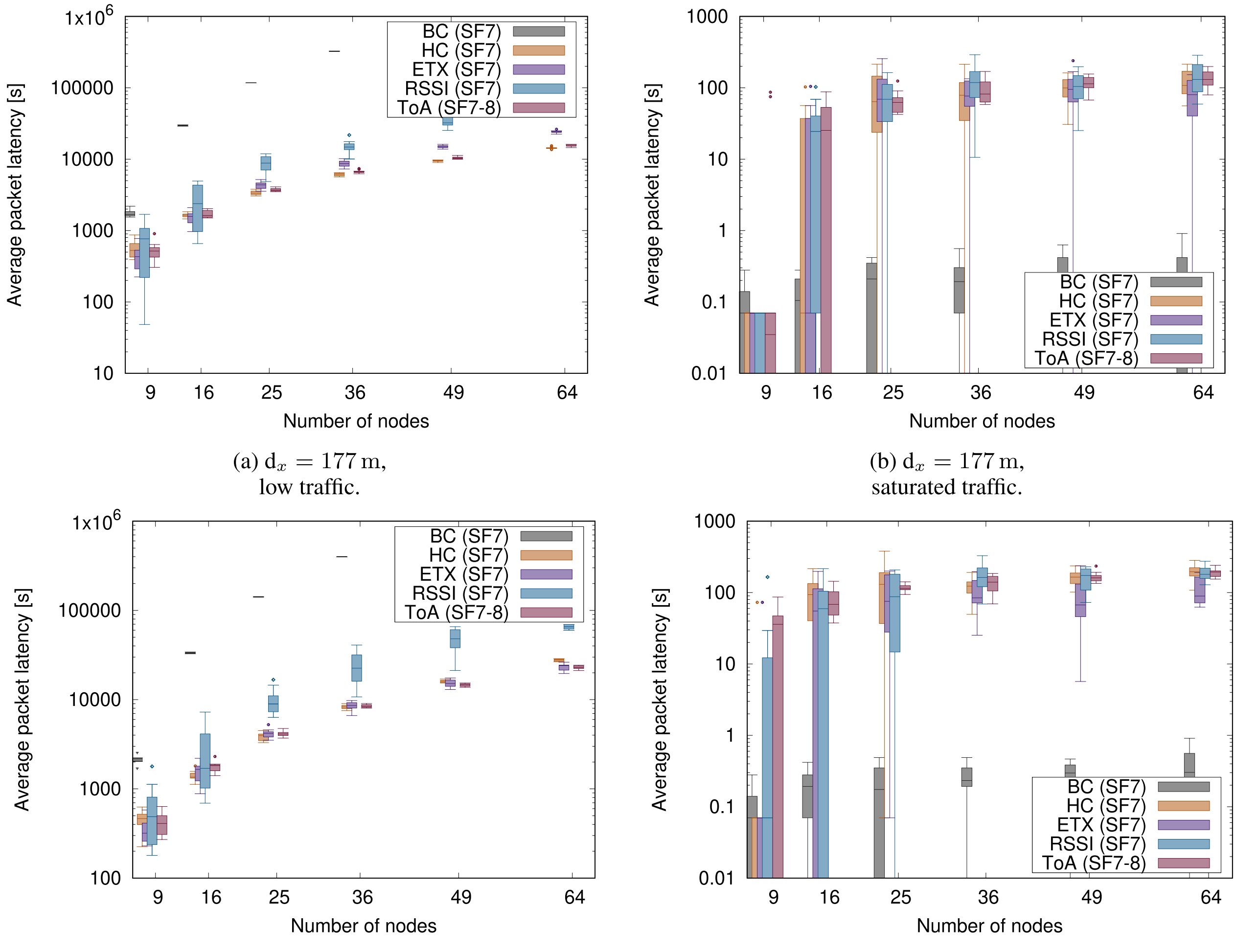


(d) Area equivalent to $d_x = 248\,\text{m}$, saturated traffic.

FIGURE 16. Average network goodput, in bps, for different amount of nodes in a random topology with an equivalent constant horizontal and vertical node spacing of 177 and 248 m, using different routing strategies. Notice the *y*-axis scale changes.

ideal conditions (i.e., continuous unidirectional transmission, no duty cycle restriction, no collisions). For a payload sized the same as the one in our simulations, the throughput ranged between 2500 bps and 100 bps, depending on the SF used (roughly, each SF step up halves the speed). However, our LoRa mesh network experiments differ from the deployments in the benchmark. First, packet collisions will occur in the absence of a scheduler to organize transmissions. Their frequency will mostly depend on the number of nodes and topologies, and the packets' egress rate. Second, the nodes will use a fraction of the available airtime to broadcast routing messages instead of data packets. Third, multi-hop communication between arbitrary pairs of nodes requires the participation of different intermediate forwarders, spending their available time to route other nodes' traffic. Therefore, the expected throughput measurements will be well below the numbers obtained in the ideal conditions benchmark.

For our experiments, we define the network's throughput as the total amount of valid data (i.e., payload) transmitted by any node *and* correctly received (i.e., no collisions) by the destination node specified in the header, measured over a given period of time. Throughput is a good KPI to understand how much data can be handled by the network. Still, it may provide an incomplete picture of its performance since packets being forwarded by different nodes may account for more throughput than single-hop communication between adjacent nodes. Also, a multi-hop packet colliding halfway to its destination would have generated throughput without reporting the benefit of end-to-end communication. To this end, we also define the network's *goodput* as the total amount of valid data transmitted end-to-end between pairs of nodes, measured over a given period of time. This magnitude provides a more precise picture of how *effective* the network is at transmitting valid data from one node to any other. To this



(a) $d_x = 177\,\mathrm{m}$, low traffic.

(b) $d_x = 177\,\mathrm{m}$, saturated traffic.

(c) $d_x = 248\,\mathrm{m}$, low traffic.

(d) $d_x = 248\,\mathrm{m}$, saturated traffic.

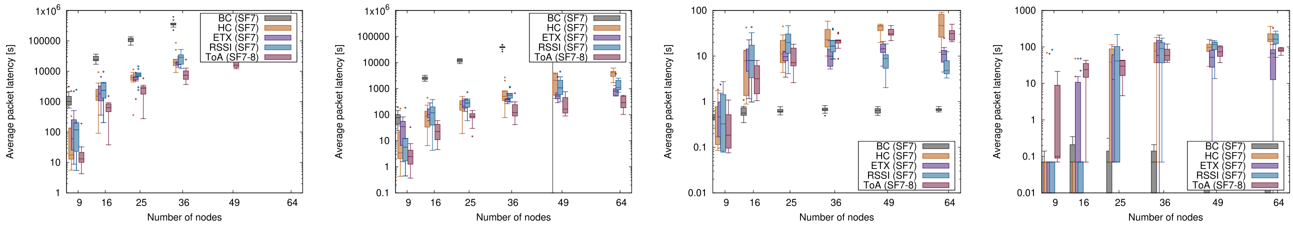
FIGURE 17. Average network latency, in bps, for different numbers of nodes in a grid topology with a constant horizontal and vertical node spacing of 177 and 248 m, using different routing strategies. Notice the *y*-axis scale change.

end, in this section, we use the goodput figures to evaluate the different routing metrics.

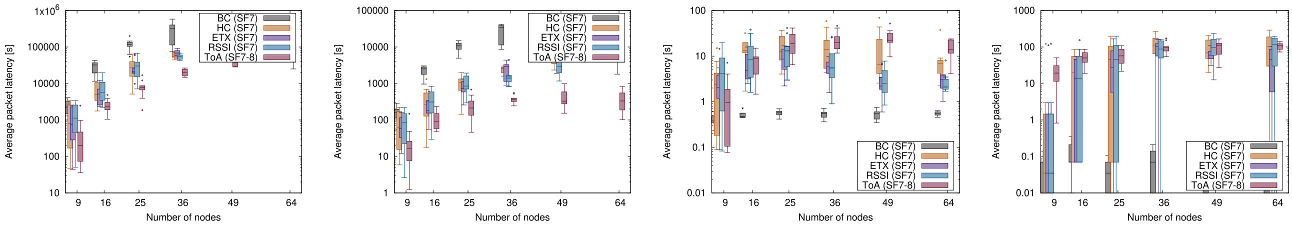
Figure 15 shows the average goodput achieved by the network, averaged to all the participating nodes. Nodes are deployed using a grid topology with a constant horizontal and vertical spacing of 177 and 248 m between them in low, medium, high, and saturation traffic scenarios. Each of the four traffic scenarios reveals a different network behavior that helps to understand the throughput KPI in a LoRa mesh network. In the low and medium traffic scenario, both single-SF and multi-SF metrics provide similar results, with HC and ToA offering comparable figures. The rest of single-SF metrics offer slightly worse performance. It is worth noting that flooding-based BC and Single Board Computer (SBC), which gave the best PDR performance in Section VI-A1, did it at the expense of significantly bad goodput results. As the traffic increases to a high volume, the multi-SF ToA offers slightly better goodput results than single-SF metrics. This metric can benefit from diagonal transmissions with the immediately higher SF when comparing the topologies with a spacing of 177 and 178 m between nodes.

In the high-traffic scenarios, where links start to become saturated, the benefits of using single-SF or multi-SF are less obvious, as some of the experiments show a relatively small advantage and others a disadvantage, suggesting that the additional complexity of multi-SF ToA may not match the elementary and regular grid topology. However, in the saturated traffic scenario (where all the nodes try to transmit packets as frequently as possible), the multi-SF ToA metric provides nodes with a mechanism to deal with some of the packet collisions. The traffic is spread on the overlaid networks with different SFs, partially desaturating the spectrum and avoiding part of the collisions. As a result, each node can still correctly transmit a few bps, while the single-SF metrics provide close to zero a goodput.

It is worth mentioning that even the best average goodput results (approx. 16 bps on SF8) are much lower than the ideal benchmark results from Section V-D. However, they are



(a) Area equivalent to $d_x = 177\,\text{m}$, low traffic.
(b) Area equivalent to $d_x = 177\,\text{m}$, medium traffic.
(c) Area equivalent to $d_x = 177\,\text{m}$, high traffic.
(d) Area equivalent to $d_x = 177\,\text{m}$, saturated traffic.



(e) Area equivalent to $d_x = 248\,\text{m}$, low traffic.
(f) Area equivalent to $d_x = 248\,\text{m}$, medium traffic.
(g) Area equivalent to $d_x = 248\,\text{m}$, high traffic.
(h) Area equivalent to $d_x = 248\,\text{m}$, saturated traffic.

FIGURE 18. Average network latency, in bps, for different amount of nodes in a random topology with an equivalent horizontal and vertical node spacing of 177 and 248 m, using different routing strategies. Notice the *y*-axis scale change.

consistent with the radio technology used and the challenges this demanding topology and traffic pattern poses.

C. LATENCY

Figure 16 shows the average goodput results achieved by the network, averaged to all the participating nodes when these are deployed randomly over an area the same size as the previous experiments. Now, the multi-SF ToA routing metric can adapt to the heterogeneity of each node, which positively impacts the performance. In most experimented cases, the best goodput figures are achieved by the ToA metric, whether we consider a low, medium, high, or saturated traffic load (medium and high omitted, minor differences). It is worth comparing the ratio between this figure and the one for the throughput (Fig. 15) to see how efficient the network is in providing end-to-end communication between pairs of nodes. In the low traffic scenario for a grid topology with nine nodes, 177 m apart, using the ToA metric with two SFs (8 to 9), nodes achieve an average throughput of 1, 85 bps (Subfig. 15a), but only deliver a goodput just below 1, 2 bps. Under good PDR conditions (i.e., few packet losses), each end-to-end packet transmission requires an average of 1.5 hops, which is in tune with the network dimensions. In any case, this ratio varies depending on the number of nodes, their spacing, the traffic load, and the routing strategy. A ratio closer to 1 may indicate that nodes use fewer hops to perform end-to-end communications (e.g., using longer-reaching SFs). At the same time, higher rates suggest that each successful end-to-end communication requires more packet transmissions, indicating either shorter-reaching SF or higher packet losses. Still, Subfigures 15d and 15h, corresponding to a saturated traffic scenario, reveal that nodes using the multi-SF ToA take advantage of LoRa's orthogonality between SFs, delivering goodput figures in the tens of bps order.

The analysis of the network's throughput and goodput performance, with different numbers of nodes, topologies, and traffic loads, shows that the data rates achieved by the nodes are much smaller than the ideal benchmark results, between two and three orders of magnitude below. Additionally, we can acknowledge that using multiple SFs simultaneously has a limited impact on the low to high traffic loads but dramatically improves throughput and goodput in traffic saturation scenarios, suggesting that, to maximize network performance, multi-SF operation would be preferred.

The last KPI analyzed for our RP is packet latency. Compared to the LoRaWAN architecture, where packet transmissions are single-hop only and have a negligible latency, latency is a crucial boundary that must be considered in multi-hop mesh networks.

Figure 17 shows the average latency experienced by packets crossing the network with a grid topology, with different node spacing (177 and 248 m). In particular, only those packets successfully delivered end-to-end are considered, not those that experienced a collision and got lost. Similarly, Figure 18 offers the results for the experiments with nodes randomly distributed over an area the same size.

The figures for both topologies show that single-SF and multi-SF metrics perform very similarly, even if some particular differences may be observed. First, flooding-based BC and SBC provide much higher latency (even three orders of magnitude or mode), as they often force links saturation and filling of the nodes' buffers, which indicates they are inferior multi-hop strategies. Second, single-SF-based metrics provide consistent results –even if some metrics have slightly better performance in some traffic conditions and other metrics in other environments–. Third, multi-SF ToA provides the best latency performance in low and medium traffic loads, especially for the random topology, again taking advantage of the nodes and links heterogeneity.

It is worth mentioning that, because of the very slow packet transmission rate of the low traffic scenarios (nodes send, at most, one packet every 100 s), messages incur very high latency numbers, in the order of tens of thousands of seconds. This is due to the number of hops needed to reach the destination and the waiting time at the intermediate nodes' buffers. As discussed in Section VII, our RP may require a modification to ensure that packets received by intermediate nodes are dispatched as soon as possible, with a different rate independent of the data generation rate.

VII. CONCLUSION

In this paper, we presented the design and evaluation of a DV RP for LoRa mesh networks, which includes a novel multi-SF ToA metric that adapts to networks with heterogeneous topologies.

Our proposed RP is designed with a minimalistic set of features to reduce memory and computing footprint. Thus, it is suitable for embedded devices with a microcontroller and a LoRa radio chip. It merges Layer 2 (L2) and Layer 3 (L3) addressing and uses a proactive broadcast mechanism for nodes to exchange routes with their neighbors, keeping the routing tables up-to-date and propagating changes over the network. Its simplicity comes at the expense of a lack of certain features, such as multicast, route discovery, node-to-node or end-to-end transmission reliability, and lower power footprint with reactive routing updates.

A novel aspect of our RP is that it takes into account LoRa's capability to transmit and receive with different SFs. This approach allows working with nodes in a mesh network using different SFs simultaneously, with packets potentially being forwarded using multiple SFs along their path. The ToA metric we present takes advantage of this feature and has proven suitable for LoRa mesh networks with heterogeneous links and topologies. For these situations where nodes have different characteristics in terms of placement, number of neighbors and distance to them, etc., our proposal achieves better PDR, goodput, and latency results than single-SF routing strategies using other metrics like HC or ETX. These contributions should be helpful in real-world deployments, where nodes are expected to operate in diverse and heterogeneous environmental conditions.

Our future work will aim to implement the protocol and the understanding gained from our simulation study into real embedded devices, leveraging existing libraries such as [29] in combination with the radio chip's CAD feature, and evaluate our DV RP and the proposed multi-SF ToA metric first in controlled laboratory environments and later seek realistic outdoors deployment.

REFERENCES

[1] C. Milarokostas, D. Tsolkas, N. Passas, and L. Merakos, "A comprehensive study on LPWANs with a focus on the potential of LoRa/LoRaWAN systems," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 825–867, 1st Quart., 2023.

[2] LoRa Alliance Technical Commitee. (2020). *LoRaWAN® L2 1.0.4 Specification*. Accessed: Apr. 22, 2024. [Online]. Available: https://resources.lora-alliance.org/technical-specifications/ts001-1-0-4-lorawan-l2-1-0-4-specification

[3] Z. Sun, H. Yang, K. Liu, Z. Yin, Z. Li, and W. Xu, "Recent advances in LoRa: A comprehensive survey," *ACM Trans. Sensor Netw.*, vol. 18, no. 4, pp. 1–44, Nov. 2022, doi: 10.1145/3543856.

[4] R. P. Centelles, F. Freitag, R. Meseguer, and L. Navarro, "Beyond the star of stars: An introduction to multihop and mesh for LoRa and LoRaWAN," *IEEE Pervasive Comput.*, vol. 20, no. 2, pp. 63–72, Apr. 2021.

[5] A. W.-L. Wong, S. L. Goh, M. K. Hasan, and S. Fattah, "Multi-hop and mesh for LoRA networks: Recent advancements, issues, and recommended applications," *ACM Comput. Surveys*, vol. 56, no. 6, pp. 1–43, Jan. 2024, doi: 10.1145/3638241.

[6] R. P. Centelles, "Towards LoRa mesh networks for the IoT," Ph.D. dissertation, Dept. Computer Architecture, Universitat Politècnica de Catalunya, Barcelona, Spain, Nov. 2021.

[7] G. Slabicki and M. Premsankar. (2018). *FLoRa—A Framework for LoRa Simulations*. Accessed: Mar. 22, 2024. [Online]. Available: https://flora.aalto.fi/

[8] O Ltd. (2020). *OMNeT++ Discrete Event Simulator*. Accessed: Apr. 22, 2024. [Online]. Available: https://omnetpp.org

[9] P. Gkotsiopoulos, D. Zorbas, and C. Douligeris, "Performance determinants in LoRA networks: A literature review," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1721–1758, 3rd Quart., 2021.

[10] J. R. Cotrim and J. H. Kleinschmidt, "LoRaWAN mesh networks: A review and classification of multihop communication," *Sensors*, vol. 20, no. 15, p. 4273, Jul. 2020. [Online]. Available: https://www.mdpi.com/1424-8220/20/15/4273

[11] A. Osorio, M. Calle, J. D. Soto, and J. E. Candelo-Becerra, "Routing in LoRaWAN: Overview and challenges," *IEEE Commun. Mag.*, vol. 58, no. 6, pp. 72–76, Jun. 2020.

[12] C. Li and Z. Cao, "LoRa networking techniques for large-scale and long-term IoT: A down-to-top survey," *ACM Comput. Surveys*, vol. 55, no. 3, pp. 1–36, Mar. 2023.

[13] LoRa Alliance Technical Commitee. (2022). *LoRaWAN® Relay Specification TS011-1.0.0*. Accessed: Apr. 22, 2024. [Online]. Available: https://resources.lora-alliance.org/technical-specifications/ts011-1-0-0-relay

[14] P. Tian, C. A. Boano, X. Ma, and J. Wei, "LoRaHop: Multi-hop support for LoRaWAN uplink and downlink messaging," *IEEE Internet Things J.*, vol. 10, no. 17, pp. 15376–15392, Sep. 2023.

[15] M. R. Islam, M. Bokhtiar-Al-Zami, B. Paul, R. Palit, J.-C. Grégoire, and S. Islam, "Performance evaluation of multi-hop LoRaWAN," *IEEE Access*, vol. 11, pp. 50929–50945, 2023.

[16] D. Lundell, A. Hedberg, C. Nyberg, and E. Fitzgerald, "A routing protocol for LoRA mesh networks," in *Proc. IEEE 19th Int. Symp. World Wireless, Mobile Multimedia Netw. (WoWMoM)*, Jun. 2018, pp. 14–19.

[17] C. Ebi, F. Schaltegger, A. Rüst, and F. Blumensaat, "Synchronous LoRA mesh network to monitor processes in underground infrastructure," *IEEE Access*, vol. 7, pp. 57663–57677, 2019.

[18] C. T. Duong and M. K. Kim, "Multi-hop linear network based on LoRa," *Adv. Sci. Technol. Lett.*, vol. 150, pp. 29–33, 2018.

[19] A. Abrardo and A. Pozzebon, "A multi-hop LoRA linear sensor network for the monitoring of underground environments: The case of the medieval aqueducts in Siena, Italy," *Sensors*, vol. 19, no. 2, p. 402, Jan. 2019.

[20] B. Sartori, S. Thielemans, M. Bezunartea, A. Braeken, and K. Steenhaut, "Enabling RPL multihop communications based on LoRa," in *Proc. IEEE 13th Int. Conf. Wireless Mobile Comput., Netw. Commun. (WiMob)*, Oct. 2017, pp. 1–8.

[21] H.-C. Lee and K.-H. Ke, "Monitoring of large-area IoT sensors using a LoRA wireless mesh network system: Design and evaluation," *IEEE Trans. Instrum. Meas.*, vol. 67, no. 9, pp. 2177–2187, Sep. 2018.

[22] G. Zhu, C.-H. Liao, T. Sakdejayont, I.-W. Lai, Y. Narusue, and H. Morikawa, "Improving the capacity of a mesh LoRA network by spreading-factor-based network clustering," *IEEE Access*, vol. 7, pp. 21584–21596, 2019.
[23] D. L. Mai and M. K. Kim, "Multi-hop LoRA network protocol with minimized latency," *Energies*, vol. 13, no. 6, p. 1368, Mar. 2020. [Online]. Available: https://www.mdpi.com/1996-1073/13/6/1368
[24] L. Prade, J. Moraes, E. de Albuquerque, D. Rosário, and C. B. Both, "Multi-radio and multi-hop LoRA communication architecture for large scale IoT deployment," *Comput. Electr. Eng.*, vol. 102, Sep. 2022, Art. no. 108242. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0045790622004773
[25] R. Berto, P. Napoletano, and M. Savi, "A LoRa-based mesh network for peer-to-peer long-range communication," *Sensors*, vol. 21, no. 13, p. 4314, Jun. 2021. [Online]. Available: https://www.mdpi.com/1424-8220/21/13/4314
[26] L. Leonardi, L. Lo Bello, and G. Patti, "MRT-LoRa: A multi-hop real-time communication protocol for industrial IoT applications over LoRA networks," *Comput. Commun.*, vol. 199, pp. 72–86, Feb. 2023, doi: 10.1016/j.comcom.2022.12.013.
[27] A. Ghosh, S. Misra, V. Udutalapally, and D. Das, "LoRaute: Routing messages in backhaul LoRA networks for underserved regions," *IEEE Internet Things J.*, vol. 10, no. 22, pp. 19964–19971, Sep. 2023.
[28] B. Arratia, E. Rosas, C. T. Calafate, J.-C. Cano, J. M. Cecilia, and P. Manzoni, "AlLoRa: Empowering environmental intelligence through an advanced LoRa-based IoT solution," *Comput. Commun.*, vol. 218, pp. 44–58, Mar. 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0140366424000641
[29] J. M. Solé, R. P. Centelles, F. Freitag, and R. Meseguer, "Implementation of a LoRA mesh library," *IEEE Access*, vol. 10, pp. 113158–113171, 2022.
[30] D. Croce, M. Gucciardo, S. Mangione, G. Santaromita, and I. Tinnirello, "Impact of LoRA imperfect orthogonality: Analysis of link-level performance," *IEEE Commun. Lett.*, vol. 22, no. 4, pp. 796–799, Apr. 2018.
[31] M. Westenberg. (2020). *Single Channel LoRaWAN Gateway*. [Online]. Available: https://github.com/things4u/ESP-1ch-Gateway
[32] S. Kim, H. Lee, and S. Jeon, "An adaptive spreading factor selection scheme for a single channel LoRA modem," *Sensors*, vol. 20, no. 4, p. 1008, Feb. 2020.
[33] D. J. Koch, M. O. Shahid, and B. Krishnaswamy, "Spreading factor detection for low-cost adaptive data rate in LoRaWAN gateways," in *Proc. 20th ACM Conf. Embedded Networked Sensor Syst.*, vol. 1. New York, NY, USA: Association for Computing Machinery, 2022, pp. 918–924.
[34] M. Slabicki, G. Premsankar, and M. Di Francesco, "Adaptive configuration of LoRA networks for dense IoT deployments," in *Proc. NOMS IEEE/IFIP Netw. Operations Manage. Symp.*, Apr. 2018, pp. 1–9.
[35] J. Petajajarvi, K. Mikhaylov, A. Roivainen, T. Hanninen, and M. Pettissalo, "On the coverage of LPWANs: Range evaluation and channel attenuation model for LoRa technology," in *Proc. 14th Int. Conf. ITS Telecommun. (ITST)*, Dec. 2015, pp. 55–59.
[36] S. Idris, T. Karunathilake, and A. Förster, "Survey and comparative study of LoRa-enabled simulators for Internet of Things and wireless sensor networks," *Sensors*, vol. 22, no. 15, p. 5546, Jul. 2022. [Online]. Available: https://www.mdpi.com/1424-8220/22/15/5546
[37] R. P. Centelles, M. Slabicki, and G. Premsankar. *FLoRaMesh—A Framework for LoRa Mesh Simulations*. Accessed: Dec. 1, 2020. [Online]. Available: https://gitlab.com/rogerpueyo/floramesh/
[38] A. Augustin, J. Yi, T. Clausen, and W. Townsley, "A study of LoRa: Long range & low power networks for the Internet of Things," *Sensors*, vol. 16, no. 9, p. 1466, Sep. 2016. [Online]. Available: https://www.mdpi.com/1424-8220/16/9/1466
[39] J. Gromes. (2020). *RadioLib—Universal Wireless Communication Library for Arduino*. [Online]. Available: https://github.com/jgromes/RadioLib

ROGER PUEYO CENTELLES received the Ph.D. degree from Universitat Politcnica de Catalunya (UPC), in 2021. He is currently a Senior Researcher with the i2CAT Foundation. His research interests include community networks, mesh and ad-hoc networks, and the IoT.

ROC MESEGUER is currently an Associate Professor with the Computer Architecture Department, Universitat Politcnica de Catalunya (UPC). His research interests include resource allocation for large-scale systems, decentralized systems applied to ambient intelligence, and bottom-up networks.

FELIX FREITAG (Member, IEEE) is currently an Associate Professor with the Computer Architecture Department, Universitat Politcnica de Catalunya (UPC). He supervised several Ph.D. theses. The full list of his publications is available at https://futur.upc.edu/FelixUrsErichFreitag. His research interests include edge computing, embedded federated learning, and LoRa mesh networks for the IoT.

ROGER BAIG VIÑAS received the Industrial Ph.D. degree in computer architecture, in 2019. In 2016, he joined the Computer Architecture Department, where he is currently an Assistant Professor. He co-founded a start-up Routek, in 2018. He was an Engineer with the Guifi.net Foundation, from 2011 to 2019, where he previously researched community networks. His current research interests include decentralized organizations and blockchain technology.

LEANDRO NAVARRO is currently a Professor with the Computer Architecture Department, UPC. He coordinated the Community-Laboratory experimental testbed and participated in several EU and national research projects. He coordinated the Erasmus Mundus Joint Doctorate in Distributed Computing (EMJD-DC). His publications are available at http://futur.upc.edu/LeandroNavarroMoldes. His research interests include scalable, community, and cooperative networks and systems.